

Unit 5

WORK, ENERGY AND POWER

54 items: 23 from official papers (114 marks), 31 from the booklet

Syllabus sections in this unit

5.1 Energy conservation

5.2 Gravitational potential energy and kinetic energy

Unit 5: Work, energy and power

1. M/J 26/P22/Q3/c(i)

- 3 Two oscillating dippers are used to create progressive transverse water waves on the surface of a lake. The waves are shown from above in Figure 3.1.

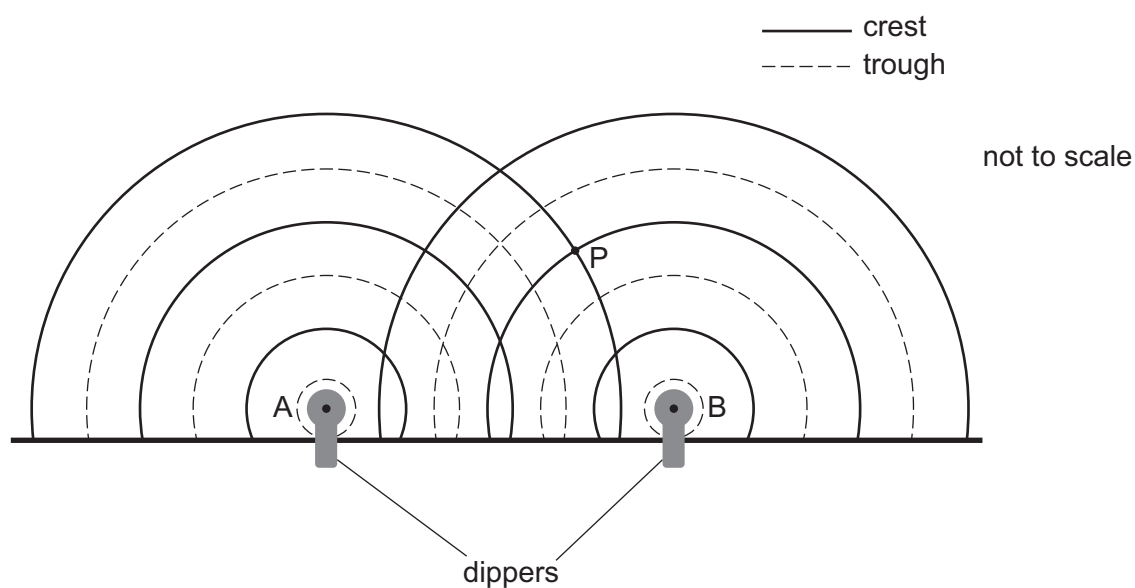


Figure 3.1

Wave crests are represented by a solid line. Wave troughs are represented by a dashed line. The waves superpose and an interference pattern is formed on the surface of the water.

The dippers oscillate in phase with each other and with a frequency of 3.0 Hz.

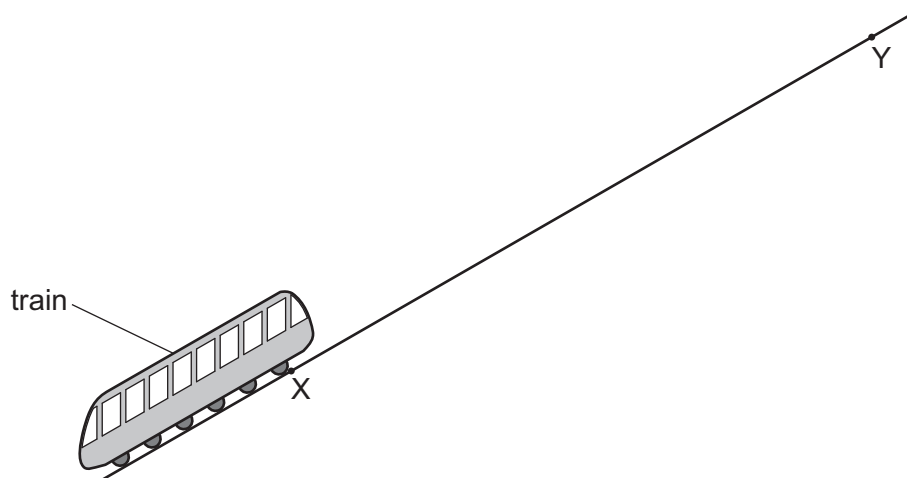
- (c) A ball of mass 0.44 kg is placed at P, where it floats on the surface of the lake. The amplitude of the vertical oscillation of the ball is 15 cm.
- (i) Calculate the difference in the gravitational potential energy ΔE of the ball between its highest and lowest points.

$\Delta E =$

J [3]

2. M/J 26/P22/Q4/a,b

- 4 An electric train of mass $1.5 \times 10^4 \text{ kg}$ travels up a slope at a constant speed as shown in Figure 4.1.

**Figure 4.1**

At point X, the train has gravitational potential energy 7.4 MJ.
 At point Y, the train has gravitational potential energy 13.0 MJ.
 The train takes a time of 52 s to move from X to Y.
 The kinetic energy of the train is $1.3 \times 10^5 \text{ J}$.

- (a) Calculate the speed of the train.

speed = ms^{-1} [3]

- (b) The train is powered by an electric motor. The useful output power of the motor is $1.1 \times 10^5 \text{ W}$.

- (i) Define power.

[1]

- (ii) Determine the resistive force F acting on the train as it moves from point X to point Y.

$F =$ N [3]

- (iii) The current in the motor is 95 A. The potential difference (p.d.) across the motor is 1400 V.

Calculate the percentage efficiency of the motor.

percentage efficiency = % [3]

3. M/J 26/P24/Q1/b(i)

- (b) A car of mass 1400 kg is initially travelling horizontally in a straight line at a constant speed of 15 ms^{-1} . The brakes are applied and the car decelerates uniformly to rest in a distance of 35 m.

Calculate:

- (i) the initial kinetic energy of the car

kinetic energy = J [2]

4. MAR 26/P22/Q3/a,b

- 3** A rollercoaster carriage of mass 280 kg travels over a raised section of track, as illustrated in Figure 3.1.

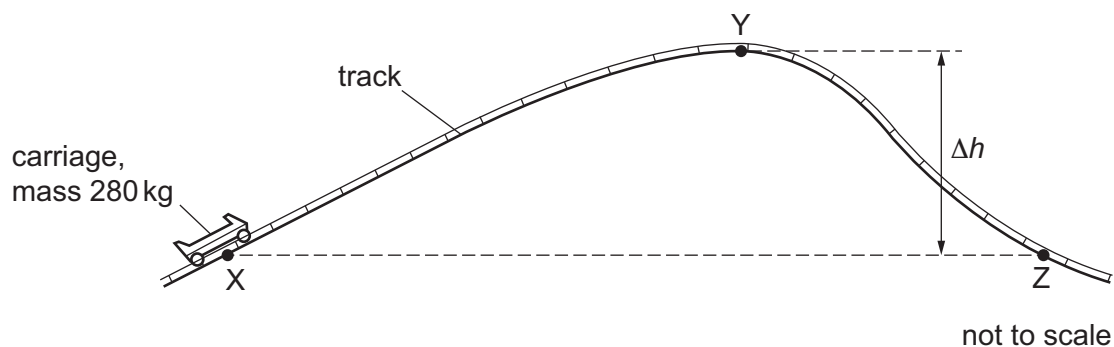


Figure 3.1

At position X, the carriage has kinetic energy 16.1 kJ. At position Y, the carriage has kinetic energy 1.3 kJ. The length of track between X and Y is 23 m.

In moving from X to Y the carriage does 5.1 kJ of work against resistive forces.

- (a) (i)** Calculate the average resistive force acting on the carriage between X and Y.

average force = N [2]

- (ii)** Calculate the change in height Δh between X and Y.

Δh = m [3]

- (b)** The carriage travels further along the track to point Z.

State and explain whether the speed of the carriage at Z is greater than, less than or the same as the speed of the carriage at X.

[1]

5. O/N 25/P21/Q1/c

(b) A rocket is launched vertically from the surface of the Earth.

Fig. 1.1 shows the variation of the velocity of the rocket with time for the first 20 s after its launch.

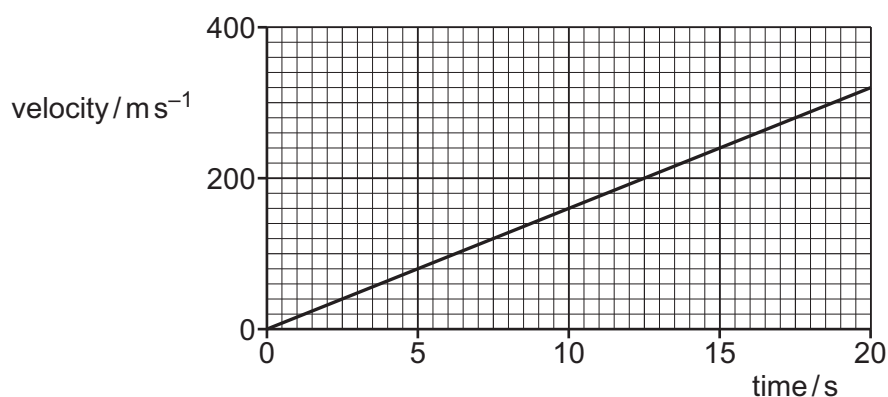


Fig. 1.1

(i) Determine the acceleration of the rocket.

$$\text{acceleration} = \quad \text{ms}^{-2} \quad [1]$$

(ii) Show that the height of the rocket above the surface of the Earth at a time of 20 s after launch is 3.2 km.

[2]

(c) The mass of the rocket in **(b)** is $2.9 \times 10^6 \text{ kg}$. Assume that this mass remains constant.

For this rocket, from launch to its height at a time of 20 s after launch:

(i) calculate the gain in gravitational potential energy ΔE_p

$$\Delta E_p = \quad \text{J} \quad [2]$$

(ii) calculate the gain in kinetic energy ΔE_k

$$\Delta E_k = \quad \text{J} \quad [2]$$

(iii) determine the average power output of the rocket engines. Assume that resistive forces are negligible.

$$\text{power} = \quad \text{W} \quad [2]$$

6. O/N 25/P24/Q3/c(i),d

- 3 A bungee jumper of mass 64 kg secures one end of an elastic rope to a bridge. The other end is attached to the bungee jumper. The jumper falls from rest from the bridge and descends into the valley below, as shown in Fig. 3.1.

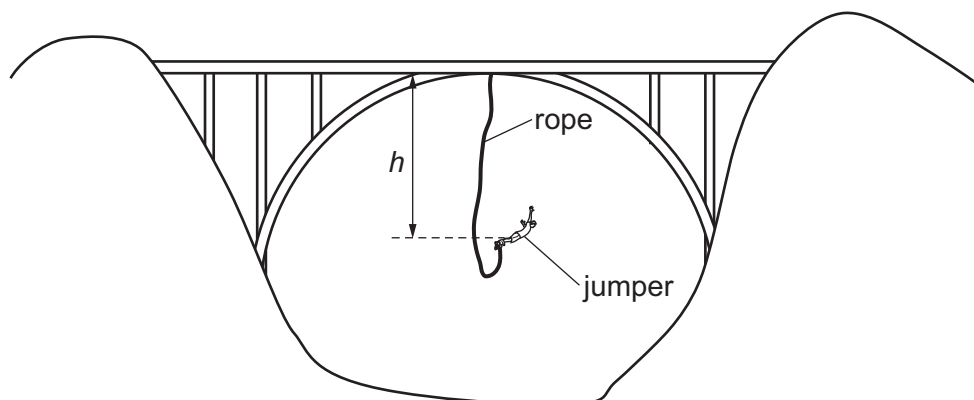


Fig. 3.1 (not to scale)

Fig. 3.2 shows the variation of the tension T in the rope with the vertical distance h of the jumper below the level of the bridge.

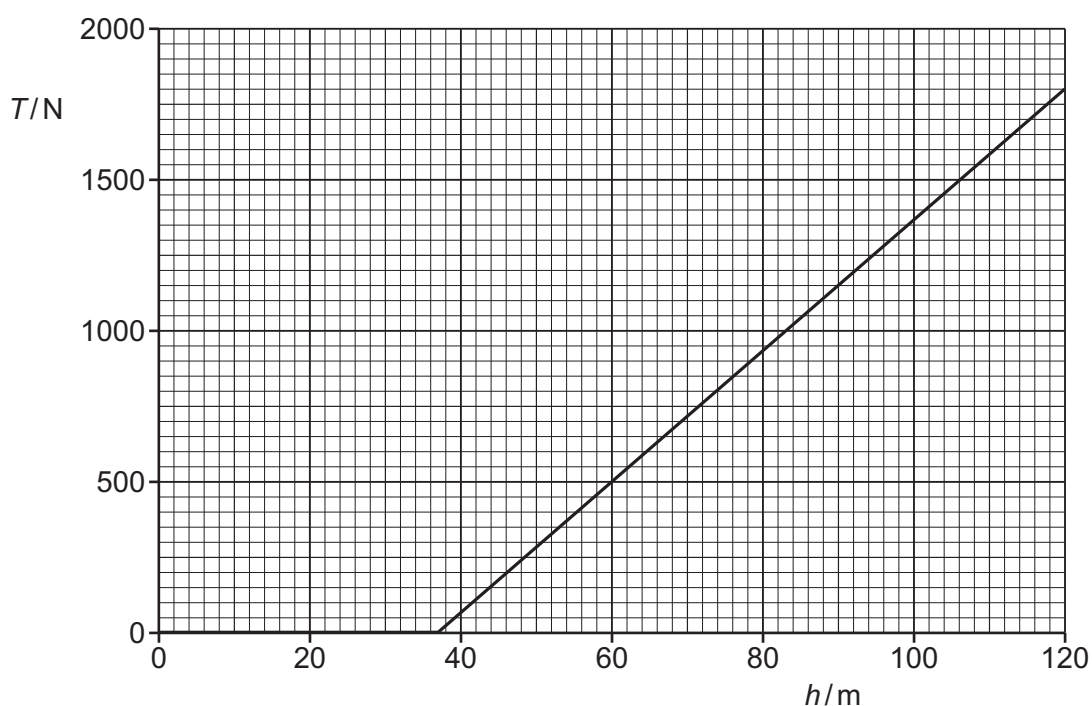


Fig. 3.2

- (c) For the position of the bungee jumper at a distance of 120 m below the bridge:
- (i) show that the loss of gravitational potential energy since leaving the bridge is 75 kJ
- (d) Explain what can be deduced from the information in (c) about the speed of the bungee jumper when at a distance of 120 m below the bridge.

[2]

[2]

7. M/J 25/P21/Q1/c(ii)

- (b) In an experiment, two objects A and B are released from the side of a building, as shown in Fig. 1.1.

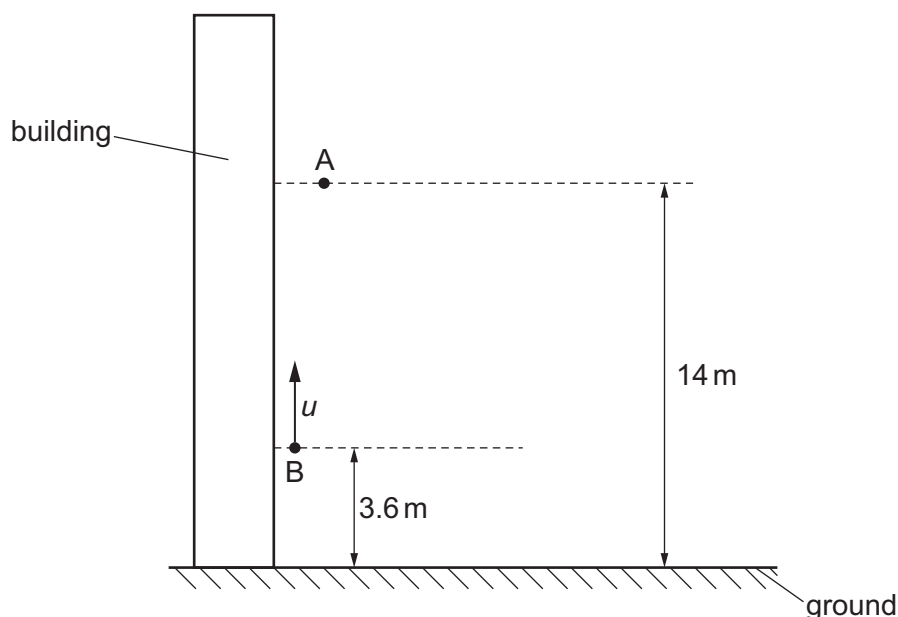


Fig. 1.1 (not to scale)

Object A is released from rest at a height of 14 m above the horizontal ground.

Object B is released with an initial upwards vertical velocity u at a height of 3.6 m above the ground.

Both objects take the same time to reach the ground and they do not collide with each other. Air resistance is negligible.

- (i) Calculate the time taken for object A to reach the ground.

time = _____ s [2]

- (ii) Use your answer in (b)(i) to calculate u .

u = _____ ms^{-1} [2]

- (c) In a second experiment, object B is released from the same height and given the same initial speed as in (b) but at a release angle θ to the vertical, as shown in Fig. 1.2.

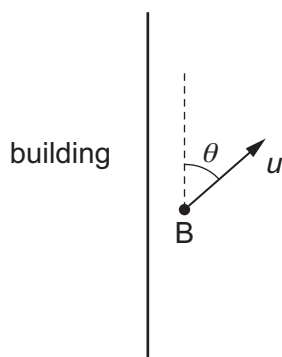


Fig. 1.2

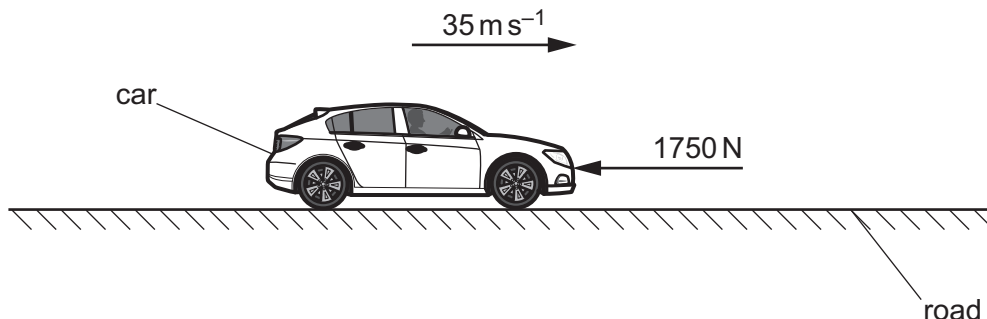
- (ii) By considering energy, state and explain the effect of the change in release angle on the speed at which B reaches the ground.

[2]

8. M/J 25/P21/Q3/a,b,c(ii)**3 (a)** Define power.

[1]

- (b)** An electric car is powered by a motor. The car is travelling at a constant speed of 35 m s^{-1} along a straight horizontal road, as shown in Fig. 3.1.

**Fig. 3.1**

There is a total resistive force of 1750 N acting on the car.

- (i)** Calculate the power transmitted to the wheels of the car by the motor.

power = W [2]

- (ii)** Calculate the useful work done by the motor when the car travels a distance of 17 km .

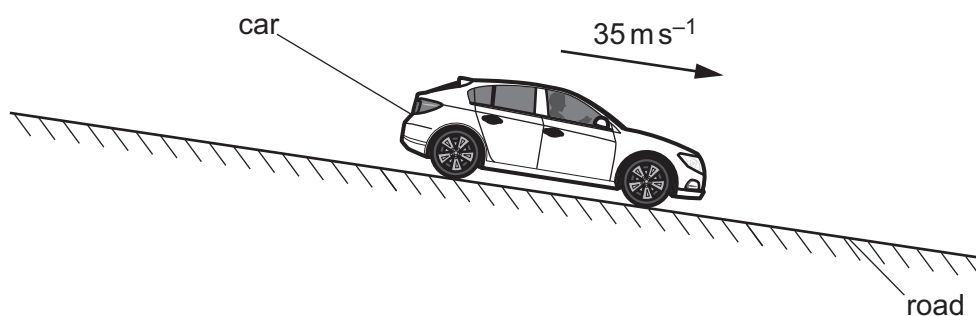
work done = J [2]

- (iii)** The potential difference (p.d.) across the motor has a constant value of 600 V and the motor has an efficiency of 85% .

Calculate the current in the motor.

current = A [3]

- (c)** The car in **(b)** now reaches a slope, as shown in Fig. 3.2.

**Fig. 3.2**

The car continues down the slope at the same speed as in **(b)**.

State and explain the effect, if any, of the slope on:

- (ii)** the current in the motor.

[1]

9. M/J 25/P22/Q1/b

- (b) A constant resultant force F acts on a car of mass m . The car moves from rest with constant acceleration a along horizontal ground. When the car has displacement s , the speed of the car is v .

- (i) Using the concept of work done on the car, show that the kinetic energy E_K of the car is given by the equation

$$E_K = \frac{1}{2}mv^2.$$

[3]

- (ii) The mass of the car is 920 kg. At time $t = 0$, the car is at rest. At time $t = 5.8$ s, its velocity is 17 m s^{-1} .

Calculate the kinetic energy of the car at time $t = 5.8$ s.

kinetic energy = J [1]

- (iii) Between time $t = 0$ and time $t = 5.8$ s, the work done against resistive forces is $4.7 \times 10^4 \text{ J}$.

Determine the average output power of the car during this time.

power = W [3]

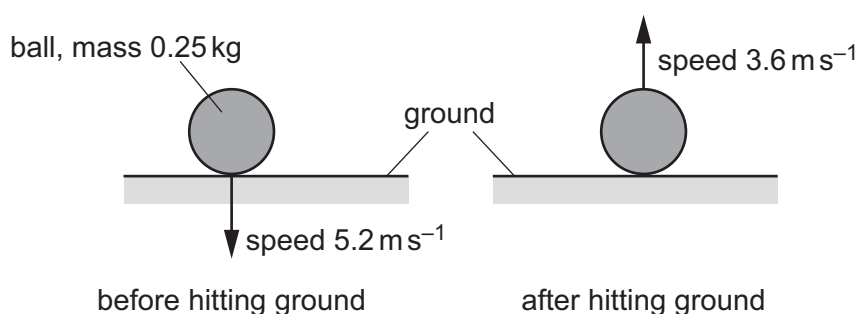
- (iv) At time $t = 5.8$ s, the speed of the car becomes constant.

State and explain whether the output power of the car is greater than, less than or the same as the output power just before $t = 5.8$ s.

[1]

10. M/J 25/P22/Q4/c

- 4 A small ball is dropped from rest from height h_1 above the ground and falls vertically downwards. The ball collides with the ground and bounces back vertically upwards, reaching a maximum height h_2 . Fig. 4.1 shows the ball just before and just after hitting the ground.

**Fig. 4.1**

The ball has mass 0.25 kg and is in contact with the ground for a time of 0.18 s. Just before the ball hits the ground, it has speed 5.2 m s^{-1} . Just after it leaves the ground, it has speed 3.6 m s^{-1} .

Air resistance acting on the ball is negligible.

- (c) Calculate the ratio $\frac{h_2}{h_1}$.

ratio = [3]

11. M/J 25/P23/Q1/c(ii)

- (b) In an experiment, two objects A and B are released from the side of a building, as shown in Fig. 1.1.

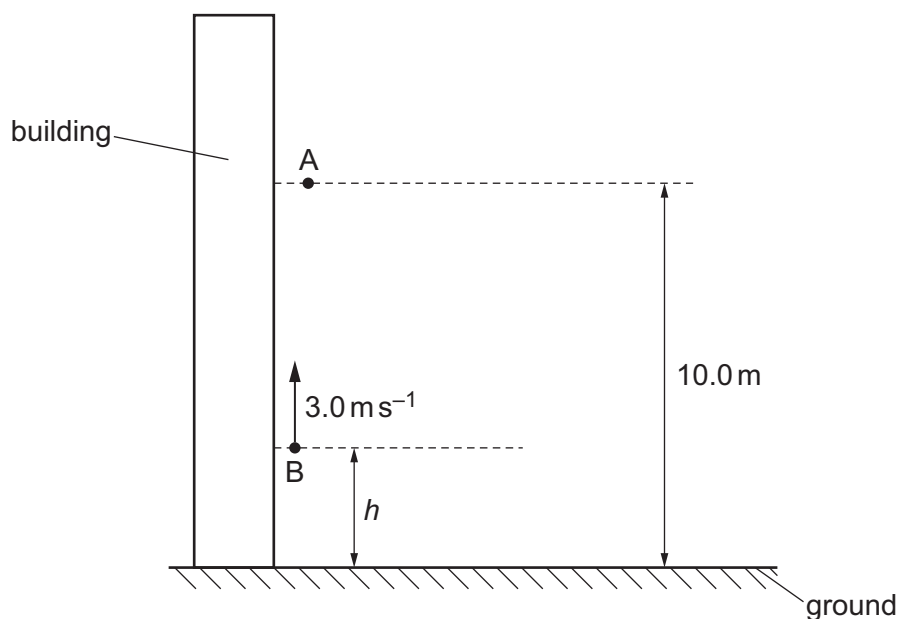


Fig. 1.1 (not to scale)

Object A is released from rest at a height of 10.0 m above horizontal ground.

Object B is released with an initial upward velocity of 3.0 m s^{-1} at a height h above the ground. Both objects take the same time to reach the ground and they do not collide with each other. Air resistance is negligible.

Calculate h .

$$h = \quad \quad \quad \text{m [3]}$$

- (c) In a second experiment, object B is released from the same height as in (b) but with a speed of 6.0 m s^{-1} at an angle of 60° to the vertical, as shown in Fig. 1.2.

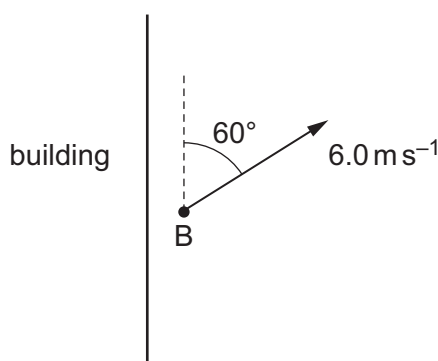


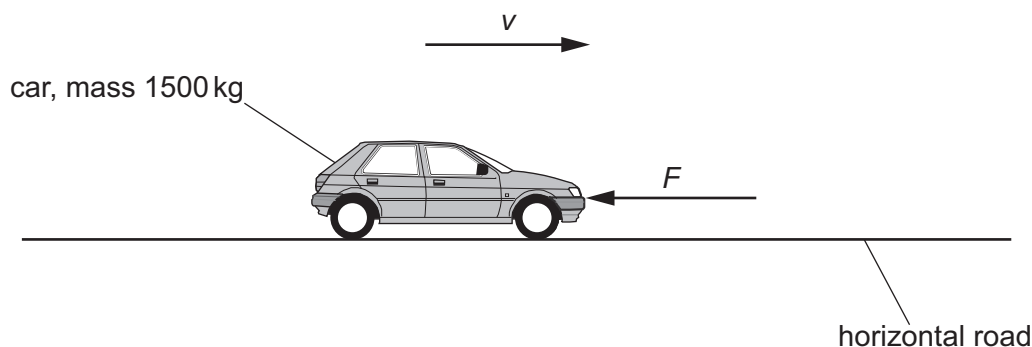
Fig. 1.2

- (ii) By considering energy, state and explain whether the speed at which object B reaches the ground is less than, the same as, or greater than in the first experiment.

[2]

12. M/J 25/P23/Q3/a,b,c(ii)

- 3 A car of mass 1500 kg is travelling along a straight horizontal road at constant velocity v . The car is subject to a total resistive force F , as shown in Fig. 3.1.

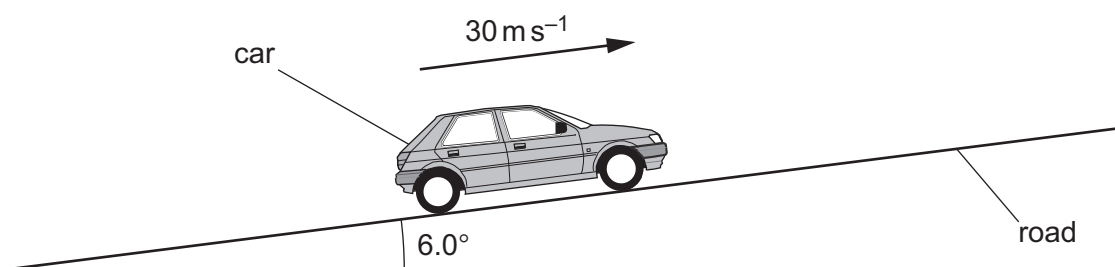
**Fig. 3.1**

- (a) Show that the power P developed by the engine in overcoming the total resistive force is given by the equation

$$P = Fv.$$

[2]

- (b) The car now moves up a slope at a constant speed of 30 m s^{-1} . The slope is at an angle to the horizontal of 6.0° , as shown in Fig. 3.2.

**Fig. 3.2**

The total resistive force acting on the car is 1600 N.

- (i) Show that the increase in gravitational potential energy of the car in a time of 1.0 s is 46 000 J.

[2]

- (ii) Use the information in (b)(i) to determine the power developed by the engine to move the car up the slope.

power = _____ W [2]

- (c) The car picks up a passenger and then continues up the slope at the same speed as in (b).

State and explain the effect, if any, that the passenger has on:

- (ii) the power developed by the engine.

[1]

13. M/J 25/P24/Q3/a,b,c

- 3** The lower end of a vertical spring is fixed to a horizontal surface, as shown in Fig. 3.1.

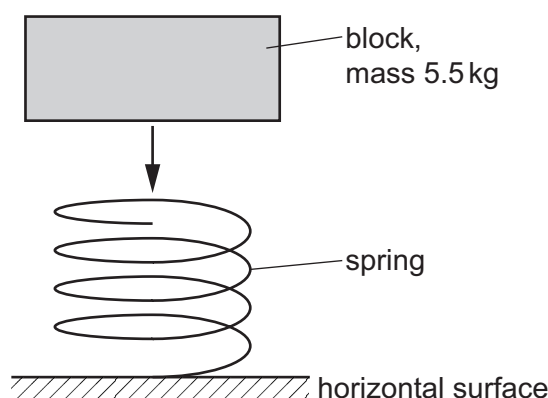


Fig. 3.1

The mass of the spring is negligible. A block of mass 5.5 kg drops vertically onto the spring and is brought to rest as the spring is compressed.

- (a)** The block has kinetic energy 110 J as it makes contact with the spring.

Calculate the speed of the block as it makes contact with the spring.

$$\text{speed} = \quad \text{ms}^{-1} \quad [2]$$

- (b)** The gravitational potential energy of the block decreases by 20 J as the spring is compressed to its maximum compression x_0 .

Show that x_0 is 0.37 m.

[2]

- (c)** Assume that, as the spring compresses, all of the energy lost by the block is converted into the elastic potential energy of the spring.

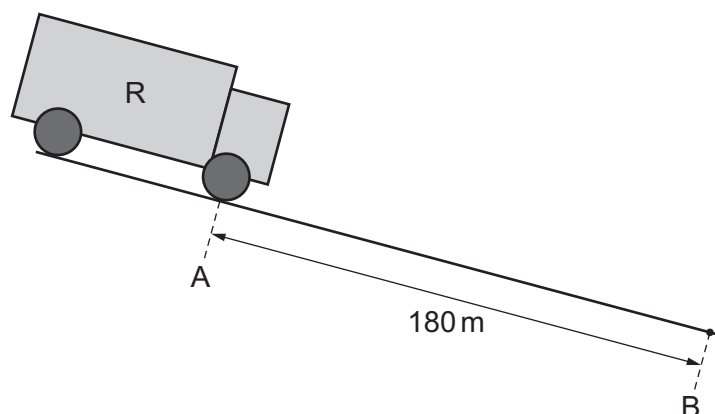
Use the data from **(a)** and **(b)** to determine the maximum elastic potential energy of the spring.

Show your working.

$$\text{maximum elastic potential energy} = \quad \text{J} \quad [1]$$

14. MAR 25/P22/Q3/a(ii)

- 3 (a) A truck R of mass 9400 kg moves with constant acceleration in a straight line down a slope, as illustrated in Fig. 3.1.

**Fig. 3.1**

At point A the speed of the truck is 13 ms^{-1} and at point B the speed of the truck is 22 ms^{-1} . A and B are a distance of 180 m apart.

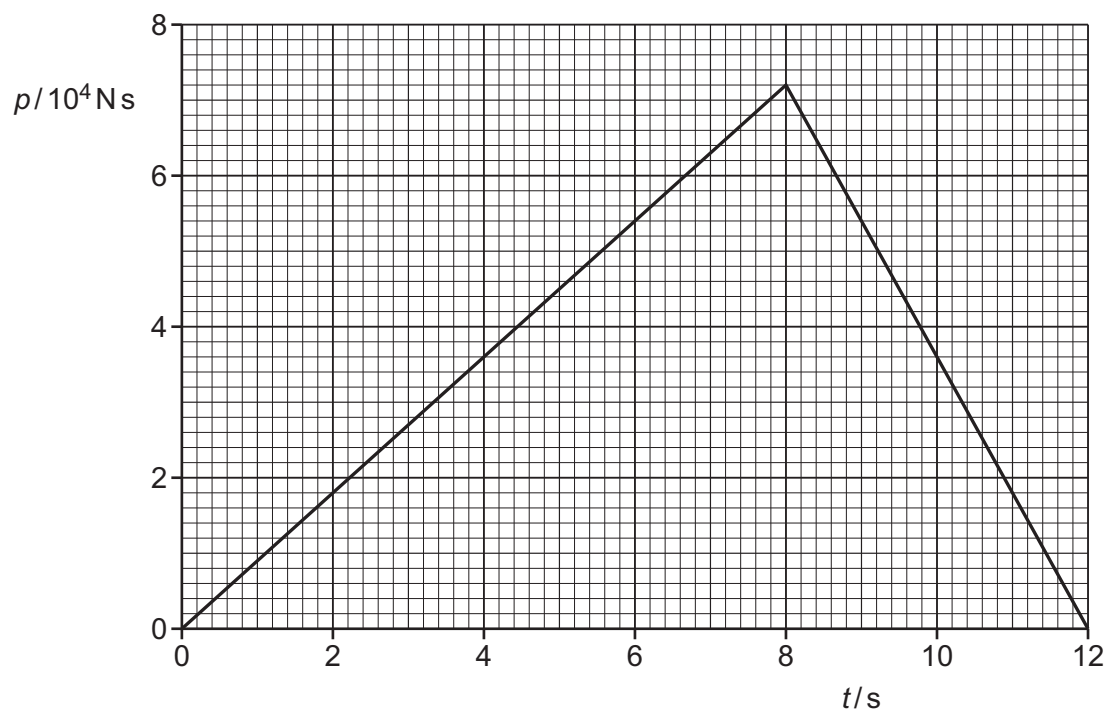
- (ii) Determine the gain in kinetic energy of the truck between A and B.

gain in kinetic energy =

J [3]

15. O/N 24/P21/Q2/b(ii)

- (b) A car of mass 1800 kg is moving in a straight line. Fig. 2.1 shows the variation with time t of the momentum p of the car.

**Fig. 2.1**

- (ii) Calculate the maximum kinetic energy of the car.

maximum kinetic energy =

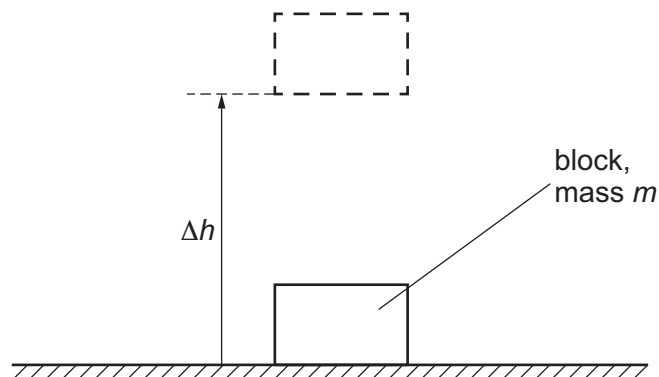
J [2]

16. O/N 24/P21/Q3/a,b,c(i,ii,iii)

- 3 (a) State what is meant by the work done by a force.

[1]

- (b) A block of mass m is raised vertically at constant speed. The vertical height gained by the block is Δh , as shown in Fig. 3.1.

**Fig. 3.1**

Derive an expression, in terms of m and Δh , for the change in gravitational potential energy ΔE_p of the block. State the meaning of any other symbols you use.

[2]

- (c) An electric motor has an input power of 900 W. The motor takes 1.0 minute to lift a load of weight 240 N at constant speed through a vertical height of 150 m. Resistive forces are negligible.

- (i) Show that the work done by the motor on the load in 1.0 minute is 36 kJ.

[1]

- (ii) Determine the useful output power of the motor.

power = W [2]

- (iii) Use your answer in (c)(ii) to determine the efficiency of the motor.

efficiency = [2]

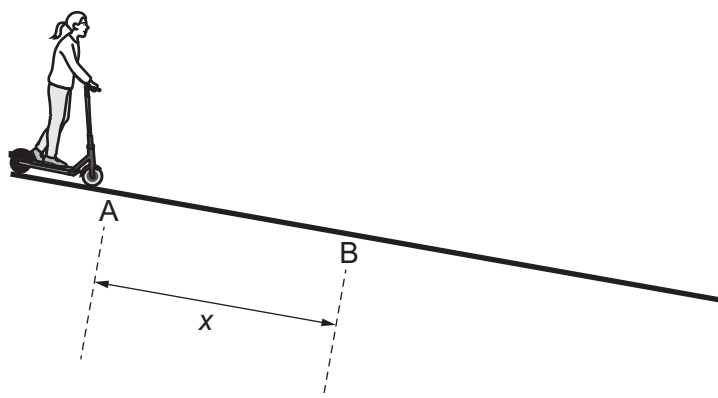
17. O/N 24/P22/Q2/d

- (b) A child stands on a scooter on horizontal ground. The combined mass of the child and the scooter is 16 kg. The child starts from rest and pushes once on the ground with her foot which causes her to accelerate. The push lasts for a time of 1.1 s. The speed of the child and the scooter after the push is 0.60 m s^{-1} .

Determine the average resultant force acting horizontally on the child and the scooter during the push.

average force = N [2]

- (c) Later, the child in (b) travels down a slope at a constant angle to the horizontal, as shown in Fig. 2.1.

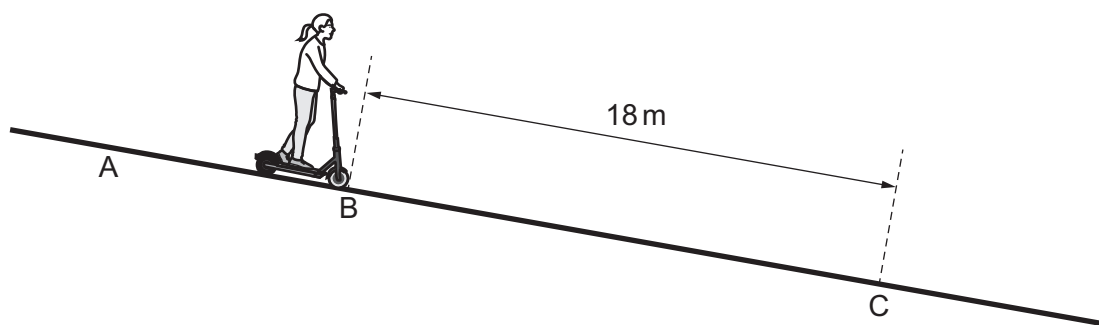
**Fig. 2.1** (not to scale)

At point A her speed is 0.60 m s^{-1} . She has a constant acceleration of 0.85 m s^{-2} parallel to the slope. After a time of 3.7 s , she reaches point B.

Calculate the distance x travelled by the child along the slope from A to B.

$x =$ _____ m [2]

- (d) At point B, the child in (c) applies the brake with a constant force to maintain a constant velocity. Point C is 18 m from point B, as shown in Fig. 2.2.

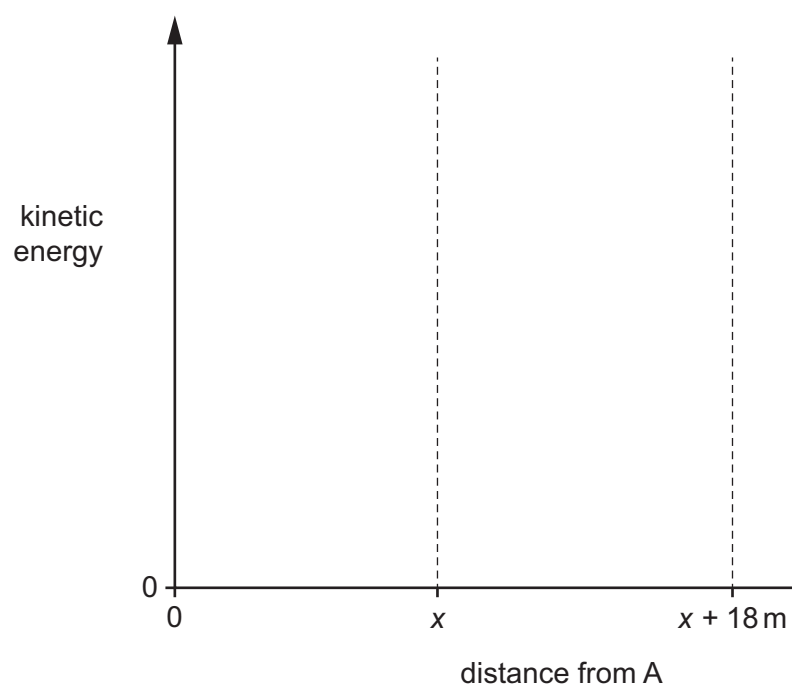
**Fig. 2.2** (not to scale)

The work done by the braking force between B and C is 250 J .

- (i) Determine the magnitude of the braking force.

force = _____ N [2]

- (ii) On Fig. 2.3, sketch the variation of the kinetic energy of the child and scooter with distance travelled from point A to point C.
Numerical values for kinetic energy are not required.

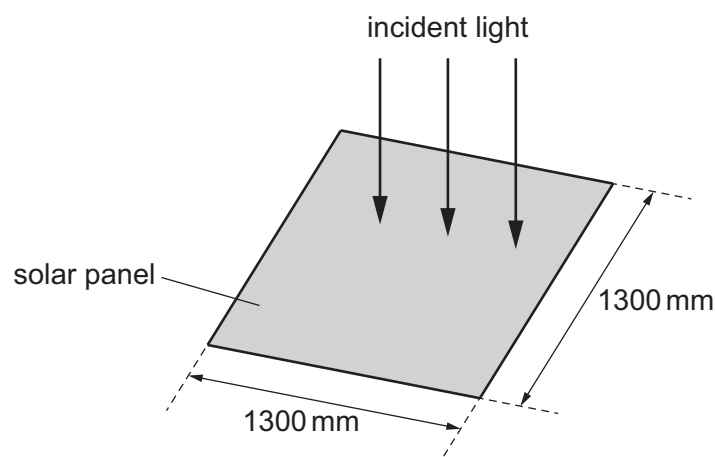
**Fig. 2.3**

[3]

18. M/J 24/P22/Q1/b

also Unit 7 (3 marks), Unit 1 (2 marks); this unit 4 marks

- (b) A square solar panel with sides of length 1300 mm is shown in Fig. 1.1.

**Fig. 1.1** (not to scale)

Light is incident normally on the solar panel.

- (i) The power of the light incident on the solar panel is 750 W.

Calculate the intensity of the light.

$$\text{intensity} = \quad \quad \quad \text{W m}^{-2} \quad [3]$$

- (ii) The percentage uncertainty in the incident power is $\pm 3\%$.
The uncertainty in the length of each side is $\pm 5 \text{ mm}$.

Calculate the percentage uncertainty in the intensity of the light.

$$\text{percentage uncertainty} = \quad \quad \quad \% \quad [2]$$



- (iii) The useful power output of the solar panel is 160 W.

Calculate the percentage efficiency of the solar panel.

$$\text{efficiency} = \quad \quad \quad \% \quad [1]$$

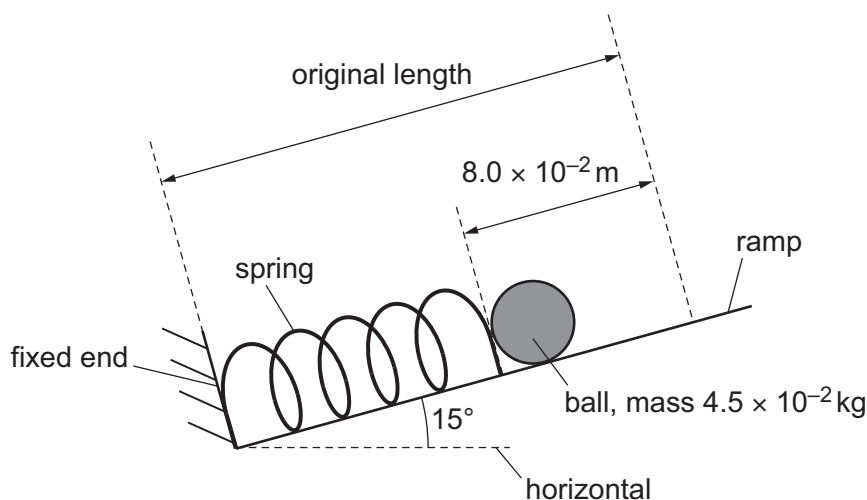
- (iv) Another square solar panel is placed so that light of the same intensity is incident normally on it. The new panel has shorter sides than the original panel. The new panel has the same power output as the original panel.

State and explain whether the efficiency of the new panel is greater than, less than or the same as the efficiency of the original panel.

[3]

19. M/J 24/P22/Q4/b

- 4 A pinball machine uses a spring to launch a small metal ball of mass $4.5 \times 10^{-2} \text{ kg}$ up a ramp. The spring is compressed by $8.0 \times 10^{-2} \text{ m}$ and held in equilibrium, as shown in Fig. 4.1.

**Fig. 4.1** (not to scale)

The ramp is at an angle of 15° to the horizontal.

- (a) The spring obeys Hooke's law and has a spring constant of 29 N m^{-1} .

Calculate the elastic potential energy in the compressed spring.

elastic potential energy = J [2]

- (b) The spring is released and expands quickly back to its original length.

- (i) Calculate the increase in gravitational potential energy of the ball when the spring returns to its original length.

increase in gravitational potential energy = J [3]



- (ii) The ball leaves the spring when the spring reaches its original length. Assume that all the elastic potential energy of the spring is transferred to the ball.

Calculate the speed of the ball as it leaves the spring.

speed = ms^{-1} [3]

20. M/J 24/P23/Q2/b(v)

(b) An object is projected horizontally at a speed of 6.0 ms^{-1} from a slope, as shown in Fig. 2.1.

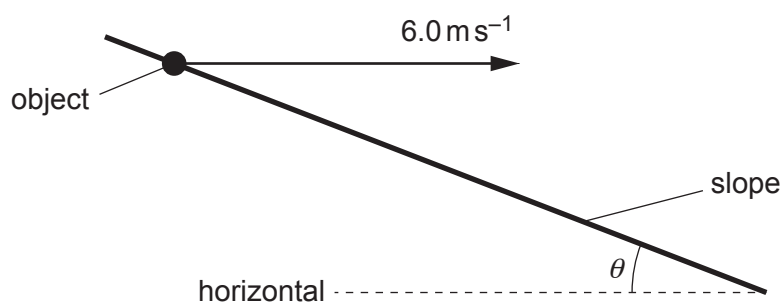


Fig. 2.1 (not to scale)

The slope is at an angle θ to the horizontal. Air resistance is negligible.

The object lands on the slope a time of 0.71 s later and stops without rolling or bouncing.

(v) By considering energy, calculate the speed of the object just before it lands.

speed = ms^{-1} [3]

21. MAR 24/P22/Q2/b

(b) An Olympic diver stands on a platform above a pool of water, as shown in Fig. 2.1.

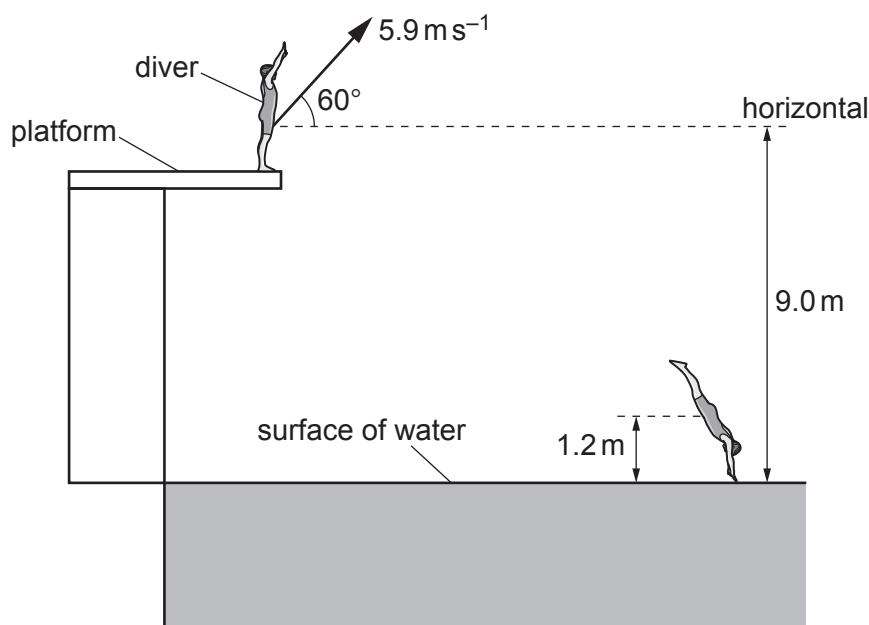


Fig. 2.1 (not to scale)

When the diver is on the platform his centre of gravity is a vertical height of 9.0 m above the surface of the water. The diver jumps from the platform with a velocity of 5.9 m s^{-1} at an angle of 60° to the horizontal.

Air resistance is negligible.

When the diver hits the surface of the water, his centre of gravity is a vertical height of 1.2 m above the surface of the water.

Calculate the speed of the diver at the instant he hits the surface of the water.

speed = ms^{-1} [3]

22. O/N 23/P22/Q3/b,c

- 3** A vertical rod is fixed to the horizontal surface of a table, as shown in Fig. 3.1.

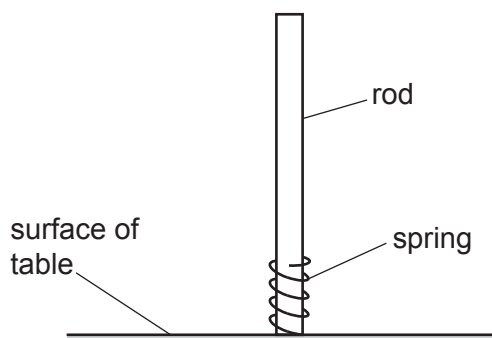


Fig. 3.1 (not to scale)

A spring of mass 7.5 g is able to slide along the full length of the rod.

The spring is first pushed against the surface of the table so that it has an initial compression of 2.1 cm. The spring is then suddenly released so that it leaves the surface of the table with a kinetic energy of 0.048 J and then moves up the rod.

Assume that the spring obeys Hooke's law and that the initial elastic potential energy of the compressed spring is equal to the kinetic energy of the spring as it leaves the surface of the table. Air resistance is negligible.

- (b)** Calculate the speed of the spring as it leaves the surface of the table.

speed = ms^{-1} [2]

- (c)** The spring rises to its maximum height up the rod from the surface of the table. This causes the gravitational potential energy of the spring to increase by 0.039 J.

- (i)** Calculate, for this movement of the spring, the increase in height of the spring after leaving the surface of the table.

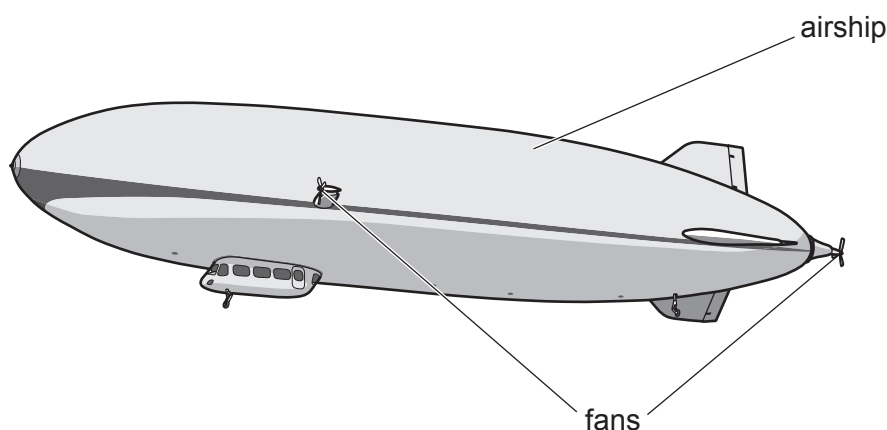
increase in height = m [2]

- (ii)** Calculate the average frictional force exerted by the rod on the spring as it rises.

average frictional force = N [2]

23. O/N 23/P23/Q3/c(iii)

- (b) Fig. 3.1 shows an airship in flight. The airship is propelled by identical fans that can be angled to control the motion of the airship.

**Fig. 3.1**

The upthrust on the airship is 93 000 N.
 The density of the surrounding air is 1.2 kg m^{-3} .

- (i) Calculate the volume of air displaced by the airship.

volume = m^3 [1]

- (ii) When fully loaded, the weight of the airship is greater than the upthrust.
 To maintain horizontal flight, the fans provide a total vertical force of $3.0 \times 10^3 \text{ N}$ upwards on the airship.

Calculate the mass of the airship.

mass = kg [2]

- (c) At a certain time, the airship in (b) is stationary. The thrust force exerted by a fan on the airship is 2800 N.

To produce this force, a mass of 64 kg of air is propelled through the blades of the fan in a time of 0.50 s. Assume that this air is initially stationary at the entrance to the fan.

Calculate:

- (iii) the total kinetic energy of this air due to its movement through the fan.

kinetic energy = J [2]

24. M/J 23/P21/Q2/c

booklet (scan + OCR)

- (a) The string is cut so that the spring extends upwards. This causes the beam to rotate and launch the block into the air. The block reaches its maximum height and then falls back to the ground.

Fig. 2.2 shows part of the path of the block in the air shortly before it hits the horizontal ground.

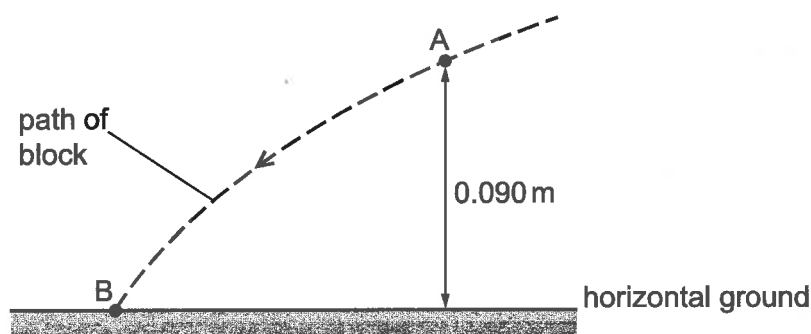


Fig. 2.2 (not to scale)

The block is at a height of 0.090 m above the ground when it passes through point A. The block has a kinetic energy of 0.044 J when it hits the ground at point B.

Air resistance is negligible.

- (i) Calculate the decrease in the gravitational potential energy of the block for its movement from A to B.

decrease in gravitational potential energy = J [2]

- (ii) Use your answer in (c)(i) and conservation of energy to determine the speed of the block at point A.

speed = ms^{-1} [3]

- (iii) By reference to the force on the block, explain why the horizontal component of the velocity of the block remains constant as it moves from A to B.

..... [1]

(iv) The block passes through point A at time t_A and arrives at point B at time t_B .

On Fig. 2.3, sketch a graph to show the variation of the magnitude of the vertical component v_y of the velocity of the block with time t from $t = t_A$ to $t = t_B$.

Numerical values of v_y are not required.

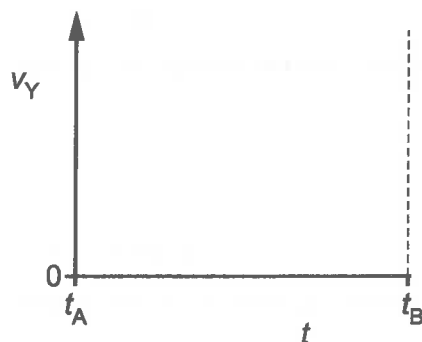


Fig. 2.3

[1]

25. MAR 23/P22/Q3

booklet (scan + OCR)

A uniform beam AB is attached by a hinge to a wall at end A, as shown in Fig. 3.1.

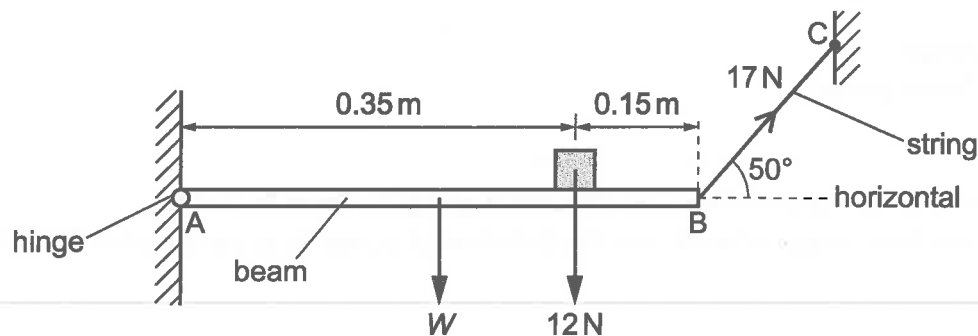


Fig. 3.1 (not to scale)

The beam has length 0.50 m and weight W . A block of weight 12 N rests on the beam at a distance of 0.15 m from end B.

The beam is held horizontal and in equilibrium by a string attached between end B and a fixed point C. The string has a tension of 17 N and is at an angle of 50° to the horizontal.

(a) State **two** conditions for an object to be in equilibrium.

1

2 [2]

(b) Show that the vertical component of the tension in the string is 13 N.

[1]

(c) By taking moments about end A, calculate the weight W of the beam.

$W = \dots\dots\dots$ N [2]

(d) Calculate the magnitude of the vertical component of the force exerted on the beam by the hinge.

force = N [1]

(e) The block is now moved closer to end A of the beam. Assume that the beam remains horizontal.

State whether this change will increase, decrease or have no effect on the horizontal component of the force exerted on the beam by the hinge.

..... [1]

26. O/N 22/P21/Q3

booklet (scan + OCR)

(a) (i) Define power.

..... [1]

(ii) Mechanical power P can be calculated using the formula $P = Fv$.

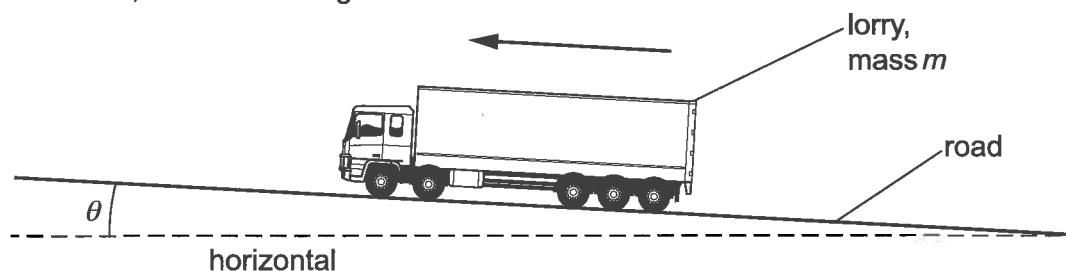
Use the concept of work and the definition of power to show how this formula is derived.

[2]

(b) The engine of a lorry provides 130 kW of power to the lorry's wheels when it is travelling at a constant speed of 25 m s^{-1} along a straight horizontal road.

Show that the resistive force opposing the forward motion of the lorry is 5200 N.

[1]

(c) The lorry in **(b)** travels up a straight section of road that is inclined at an angle θ to the horizontal, as shown in Fig. 3.1.**Fig. 3.1** (not to scale)The lorry has mass m and the acceleration of free fall is g .**(i)** Determine an expression, in terms of m , g and θ , for the component of the weight of the lorry that acts parallel to the surface of the road.

[1]

(ii) The total resistive force remains unchanged at 5200 N and the engine now provides greater power to maintain the speed of 25 m s^{-1} . The total mass m of the lorry is 36 000 kg.The angle θ is 1.4° .

Determine the power, in kW, now provided by the engine.

power = kW [3]

27. M/J 22/P22/Q3

booklet (scan + OCR)

A man standing on a wall throws a small ball vertically upwards with a velocity of 5.6 m s^{-1} . The ball leaves his hand when it is at a height of 3.1 m above the ground, as shown in Fig. 3.1.

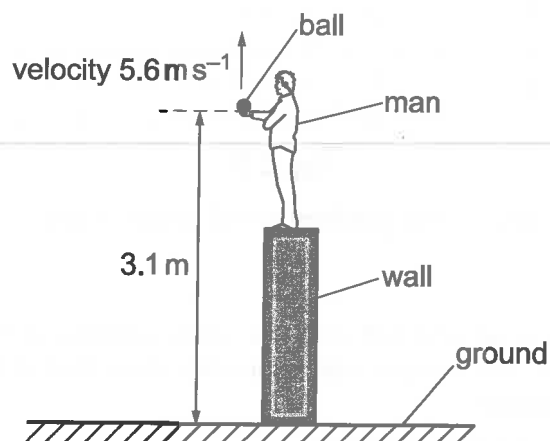


Fig. 3.1 (not to scale)

Assume that air resistance is negligible.

(a) Show that the ball reaches a maximum height above the ground of 4.7 m .

[2]

- (b) The man does not catch the ball as it falls.

Calculate the time taken for the ball to fall from its maximum height to the ground.

time taken = s [2]

- (c) The ball leaves the man's hand at time $t = 0$ and hits the ground at time $t = T$.

On Fig. 3.2, sketch a graph to show the variation of the velocity v of the ball with time t from $t = 0$ to $t = T$. Numerical values of v and t are not required. Assume that v is positive in the upward direction.

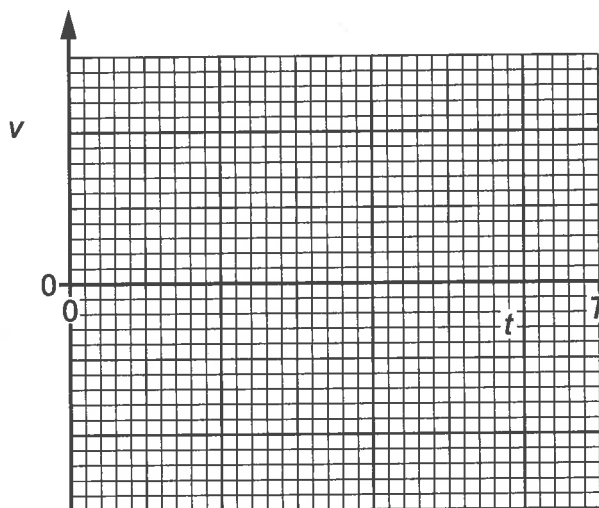


Fig. 3.2

[3]

- (d) State what is represented by the gradient of the graph in (c).

..... [1]

- (e) The man now throws a second ball with the same velocity and from the same height as the first ball. The mass of the second ball is greater than that of the first ball. Assume that air resistance is still negligible.

For the first and second balls, compare:

- (i) the magnitudes of their accelerations

..... [1]

- (ii) the speeds with which they hit the ground.

..... [1]

28. M/J 22/P21/Q3

booklet (scan + OCR)

- (a) (i) Define power.

..... [1]

- (ii) A force F takes time t to move an object through a displacement x at constant velocity v in the direction of the force. The work done by the force is W .

Use the definition of power to show that the power P transferred by the force is given by

$$P = Fv.$$

[2]

- (b) A block is pulled up a slope by a wire attached to a motor, as shown in Fig. 3.1.

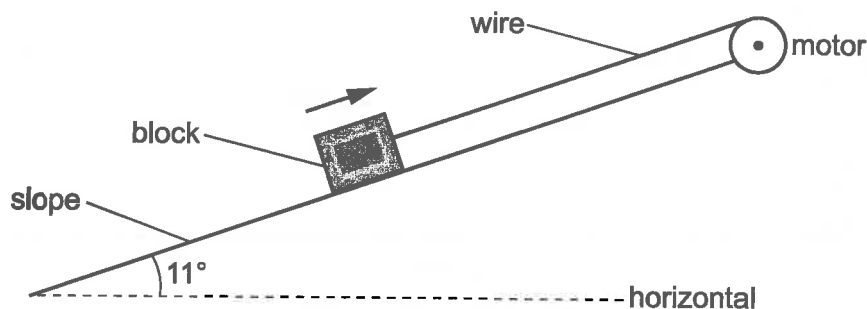


Fig. 3.1 (not to scale)

The useful power output of the motor is 56 W. The block has a weight of 430 N and travels with constant velocity along the slope at an angle of 11° to the horizontal.

Assume that there are no resistive forces opposing the motion of the block.

- (i) Calculate the tension T in the wire.

$T = \dots\dots\dots$ N [2]

- (ii) Calculate the speed of the block.

speed = $\dots\dots\dots$ ms^{-1} [1]

- (iii) The rate of increase of gravitational potential energy of the block is equal to the useful power output of the motor. One of the reasons for this is that there is no work done against resistive forces.

By considering the motion of the block, state another reason for this.

$\dots\dots\dots$ [1]

- (iv) The motor has an efficiency of 80%.

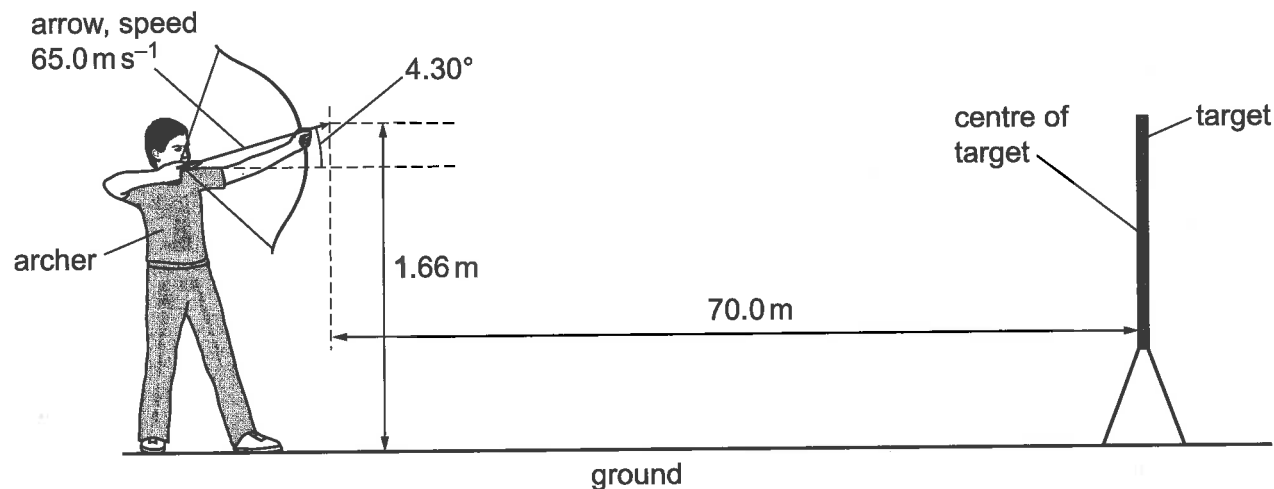
Calculate the time taken for an input energy of 1.2 kJ to be supplied to the motor.

time taken = $\dots\dots\dots$ s [2]

29. M/J 22/P23/Q2

booklet (scan + OCR)

An archer releases an arrow towards a target at a velocity of 65.0 m s^{-1} at an angle of 4.30° above the horizontal, as shown in Fig. 2.1.

**Fig. 2.1** (not to scale)

When released, the tip of the arrow is a horizontal distance of 70.0 m from the target and 1.66 m above the horizontal ground.

The arrow hits the centre of the target.

Assume that air resistance is negligible and that all the mass of the arrow is at its tip.

(a) Show that the time taken for the arrow to reach the target is 1.08 s .

[2]

(b) Calculate the height of the centre of the target above the ground.

height above ground = m [3]

(c) By considering energy changes, state and explain how the final kinetic energy of the arrow as it hits the target compares with its initial kinetic energy immediately after release. A numerical calculation is not required.

.....
 [2]

30. M/J 22/P23/Q4

booklet (scan + OCR)

(a) A mass m moves a vertical distance Δh in a uniform gravitational field and gains gravitational potential energy ΔE_p . The acceleration of free fall is g .

Use the concept of work done to show that

$$\Delta E_p = mg\Delta h.$$

[2]

- (b) A 0.60 kg mass is attached to a string which is wrapped around the wheel of a generator, as shown in Fig. 4.1.

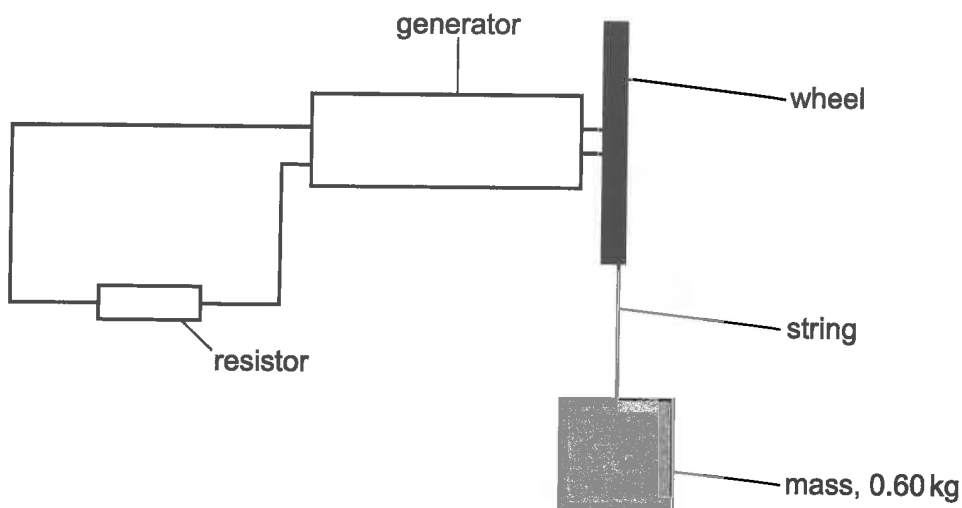


Fig. 4.1

The mass is held stationary above the floor. When released, the mass initially accelerates and then falls at a steady speed and spins the wheel. The generator causes a current in a resistor. Air resistance is negligible.

State the main energy change when the mass is falling at a steady speed.

..... energy to energy. [1]

- (c) When falling at a steady speed, the mass in (b) falls through a vertical distance of 1.4 m in a time of 4.0 s. This causes a current of 90 mA in the resistor. The resistance of the resistor is $47\ \Omega$.

Calculate:

- (i) the rate of work done by the falling mass

rate of work done = W [2]

(ii) the power dissipated in the resistor

power = W [2]

(iii) the efficiency of the generator.

efficiency = [2]

31. MAR 22/P22/Q4

booklet (scan + OCR)

A child moves down a long slide, as shown in Fig. 4.1.

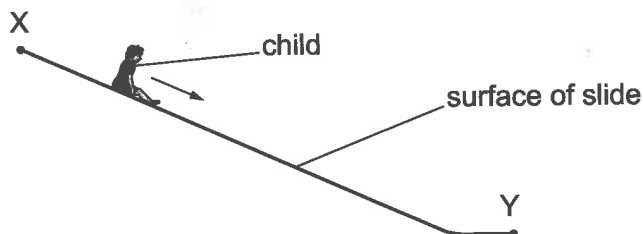


Fig. 4.1 (not to scale)

The child moves from rest at the top end X of the slide. An average resistive force of 76 N opposes the motion of the child as they move to the lower end Y of the slide. The kinetic energy of the child at Y is 300 J. The decrease in gravitational potential energy of the child as it moves from X to Y is 3200 J.

(a) Determine the ratio

$$\frac{\text{kinetic energy of the child at Y when the resistive force is 76 N}}{\text{kinetic energy of the child at Y if there is no resistive force}}$$

ratio = [1]

(b) Use the answer in (a) to calculate the ratio

$$\frac{\text{speed of the child at Y when the resistive force is 76 N}}{\text{speed of the child at Y if there is no resistive force}}$$

ratio = [2]

(c) Calculate the length of the slide from X to Y.

length = m [2]

(d) At end Y of the slide, the child is brought to rest by a board, as shown in Fig. 4.2.

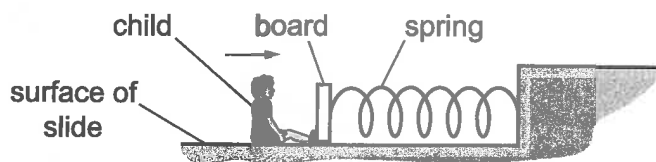


Fig. 4.2 (not to scale)

A spring connects the board to a fixed point. The spring obeys Hooke's law and has a spring constant of 63 N m^{-1} . The child hits the board so that it moves to the right and compresses the spring. The speed of the child becomes zero when the elastic potential energy of the spring has increased to its maximum value of 140 J .

(i) Calculate the maximum compression of the spring.

maximum compression = m [2]

(ii) Calculate the percentage efficiency of the transfer of the kinetic energy of the child to the elastic potential energy of the spring.

percentage efficiency = % [1]

(iii) The maximum compression of the spring is x_0 . On Fig. 4.3, sketch a graph to show the variation of the elastic potential energy of the spring with its compression x from $x = 0$ to $x = x_0$. Numerical values are not required.

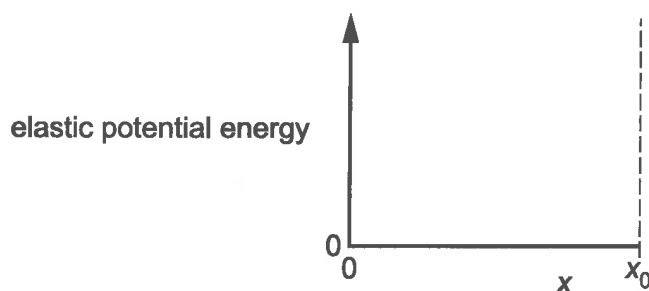


Fig. 4.3

[2]

32. O/N 21/P22/Q3

booklet (scan + OCR)

(a) Define *power*.

..... [1]

(b) A car of mass 1700 kg moves in a straight line along a slope that is at an angle θ to the horizontal, as shown in Fig. 3.1.

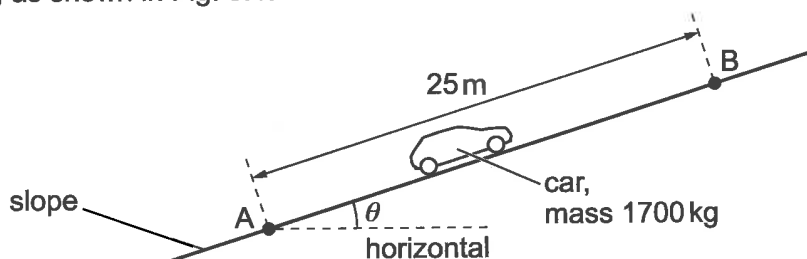


Fig. 3.1 (not to scale)

The car moves at constant velocity for a distance of 25 m from point A to point B.

Air resistance and friction provide a total resistive force of 440 N that opposes the motion of the car.

For the movement of the car from A to B:

(i) state the change in the kinetic energy

change in kinetic energy = J [1]

(ii) calculate the work done against the total resistive force.

work done = J [1]

(c) The movement of the car in (b) from A to B causes its gravitational potential energy to increase by 4.8×10^4 J.

Calculate:

(i) the increase in vertical height h of the car for its movement from A to B

$h =$ m [2]

(ii) angle θ .

$\theta =$ $^\circ$ [1]

(d) The engine of the car in (b) produces an output power of 1.7×10^4 W to move the car along the slope.

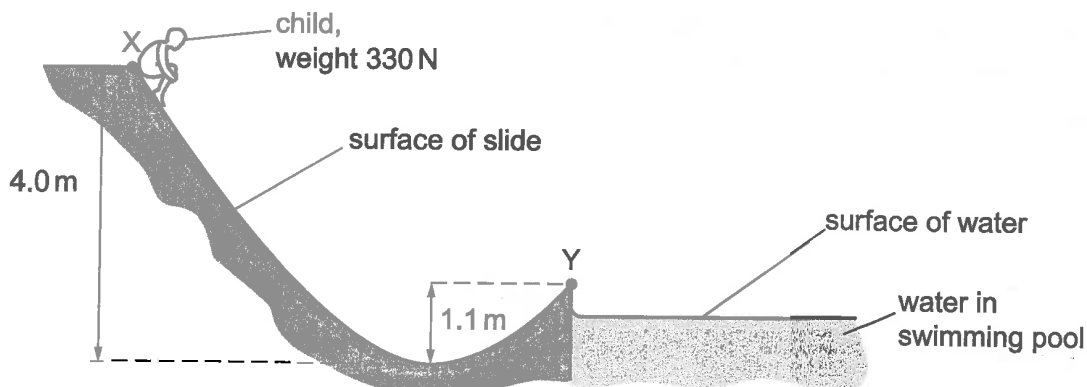
Calculate the time taken for the car to move from A to B.

time = s [2]

33. M/J 21/P22/Q3

booklet (scan + OCR)

A child of weight 330 N is at point X at the top of a slide. The slide is at the edge of a swimming pool, as shown in Fig. 3.1.

**Fig. 3.1** (not to scale)

The child moves from rest to the lowest point of the slide that is a vertical distance of 4.0 m below X. The child continues moving towards point Y which is at the end of the slide and a vertical distance of 1.1 m above the lowest point. The kinetic energy of the child at Y is 540 J .

(a) Calculate the difference in the gravitational potential energy of the child at points X and Y.

difference in gravitational potential energy = J [2]

(b) An average frictional force of 52 N acts on the child when moving from X to Y.

By considering changes of energy, determine the distance moved by the child from X to Y.

distance moved = m [2]

- (c) The child leaves the slide at point Y with a velocity that is at an angle of 41° to the horizontal. The path of the child through the air is shown in Fig. 3.2.

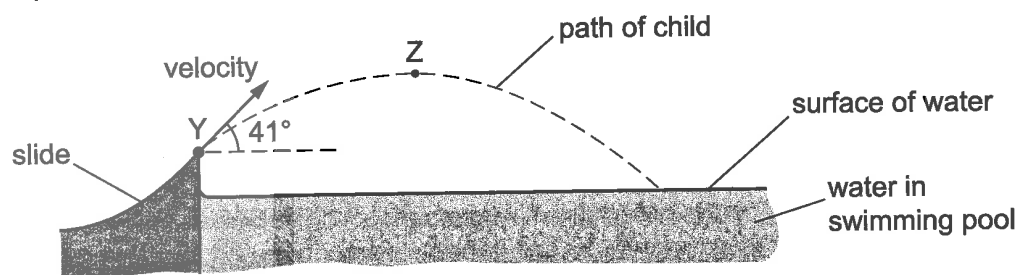


Fig. 3.2 (not to scale)

Point Z is the highest point on the path of the child through the air.
Assume that air resistance is negligible.

Calculate the speed of the child at:

- (i) point Y

speed = ms^{-1} [2]

- (ii) point Z.

speed = ms^{-1} [2]

34. MAR 21/P22/Q2

booklet (scan + OCR)

(a) State what is meant by *work done*.

..... [1]

(b) A beach ball is released from a balcony at the top of a tall building. The ball falls vertically from rest and reaches a constant (terminal) velocity. The gravitational potential energy of the ball decreases by 60 J as it falls from the balcony to the ground. The ball hits the ground with speed 16 m s^{-1} and kinetic energy 23 J.

(i) Show that the mass of the ball is 0.18 kg.

[2]

(ii) Calculate the height of the balcony above the ground.

height = m [2]

(iii) Determine the average resistive force acting on the ball as it falls from the balcony to the ground.

average resistive force = N [2]

(c) State and explain the variation, if any, in the magnitude of the acceleration of the ball in (b) during the time interval when the ball is moving downwards **before** it reaches constant (terminal) velocity.

.....

.....

..... [3]

35. M/J 20/P22/Q3

booklet (scan + OCR)

(a) Explain what is meant by *work done*.

..... [1]

(b) A ball of mass 0.42 kg is dropped from the top of a building. The ball falls from rest through a vertical distance of 78 m to the ground. Air resistance is significant so that the ball reaches constant (terminal) velocity before hitting the ground. The ball hits the ground with a speed of 23 ms^{-1} .

(i) Calculate, for the ball falling from the top of the building to the ground:

1. the decrease in gravitational potential energy

decrease in gravitational potential energy = J [2]

2. the increase in kinetic energy.

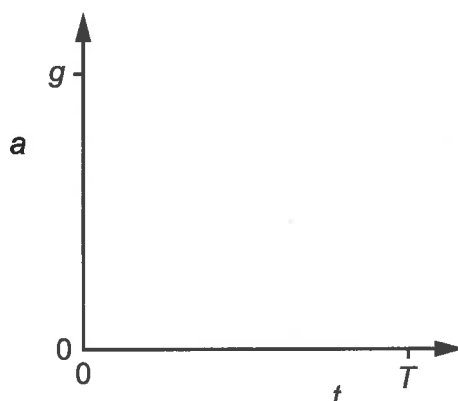
increase in kinetic energy = J [2]

(ii) Use your answers in **(b)(i)** to determine the average resistive force acting on the ball as it falls from the top of the building to the ground.

average resistive force = N [2]

(c) The ball in **(b)** is dropped at time $t = 0$ and hits the ground at time $t = T$. The acceleration of free fall is g .

On Fig. 3.1, sketch a line to show the variation of the acceleration a of the ball with time t from time $t = 0$ to $t = T$.

**Fig. 3.1**

[2]

36. MAR 20/P22/Q3

booklet (scan + OCR)

(a) State what is meant by *work done*.

..... [1]

(b) A skier is pulled along horizontal ground by a wire attached to a kite, as shown in Fig. 3.1.

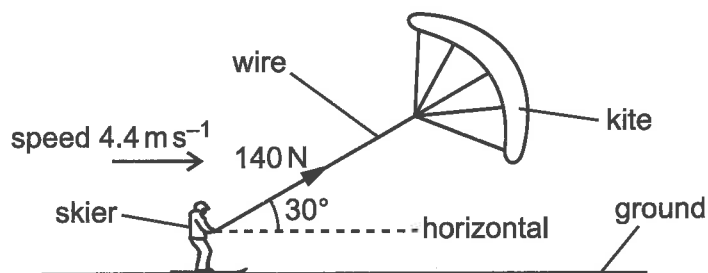


Fig. 3.1 (not to scale)

The skier moves in a straight line along the ground with a constant speed of 4.4 m s^{-1} . The wire is at an angle of 30° to the horizontal. The tension in the wire is 140 N .

(i) Calculate the work done by the tension to move the skier for a time of 30 s .

work done = J [3]

- (ii) The weight of the skier is 860 N. The vertical component of the tension in the wire and the weight of the skier combine so that the skier exerts a downward pressure on the ground of 2400 Pa.

Determine the total area of the skis in contact with the ground.

area = m² [3]

- (iii) The wire attached to the kite is uniform. The stress in the wire is 9.6×10^6 Pa.

Calculate the diameter of the wire.

diameter = m [2]

- (c) The variation with extension x of the tension F in the wire in (b) is shown in Fig. 3.2.

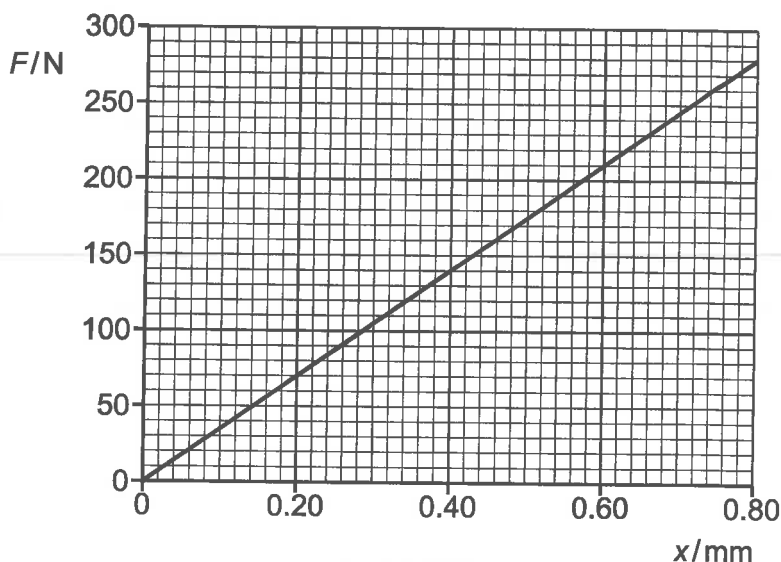


Fig. 3.2

A gust of wind increases the tension in the wire from 140 N to 210 N.

Calculate the change in the strain energy stored in the wire.

change in strain energy = J [3]

37. O/N 19/P23/Q3

booklet (scan + OCR)

May be outside the 2025–27 syllabus

(a) State the property of an object that experiences a force when the object is placed in:

(i) a gravitational field

..... [1]

(ii) an electric field.

..... [1]

(b) A potential difference of $1.2 \times 10^3 \text{ V}$ is applied between a pair of horizontal metal plates in a vacuum, as shown in Fig. 3.1.

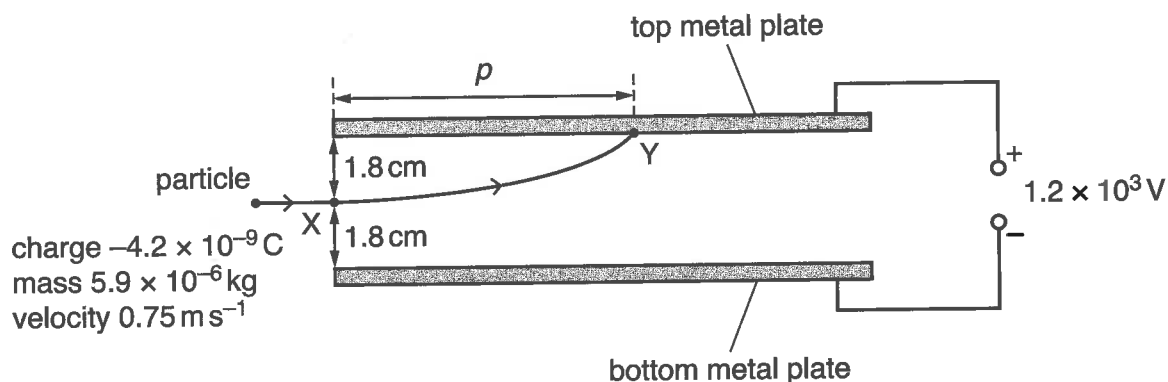


Fig. 3.1 (not to scale)

The separation of the plates is 3.6 cm. The electric field between the plates is uniform.

A particle of mass $5.9 \times 10^{-6} \text{ kg}$ and charge $-4.2 \times 10^{-9} \text{ C}$ enters the field at point X with a horizontal velocity of 0.75 m s^{-1} along a line midway between the two plates.

The particle is deflected by the field and hits the top plate at point Y.

(i) Calculate the magnitude of the electric force acting on the particle in the field.

electric force = N [3]

(ii) By considering the resultant vertical force acting on the particle, show that the acceleration of the particle in the electric and gravitational fields is 14 m s^{-2} .

[4]

(iii) Determine:

- the time taken for the particle to move from X to Y

time taken = s [2]

- the distance p of point Y from the left-hand edge of the top plate.

$p =$ m [1]

38. M/J 19/P23/Q2

booklet (scan + OCR)

- (a)** A resultant force F moves an object of mass m through distance s in a straight line. The force gives the object an acceleration a so that its speed changes from initial speed u to final speed v .

(i) State an expression for:

- the work W done by the force, in terms of a , m and s

$W =$ [1]

- the distance s , in terms of a , u and v .

$s =$ [1]

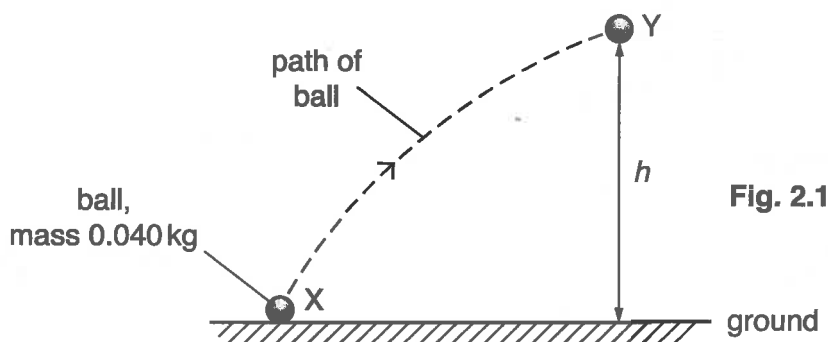
(ii) Use your answers in (i) to show that the kinetic energy of the object is given by

$$\text{kinetic energy} = \frac{1}{2} \times \text{mass} \times (\text{speed})^2.$$

Explain your working.

[2]

- (b)** A ball of mass 0.040 kg is projected into the air from horizontal ground, as illustrated in Fig. 2.1.



The ball is launched from a point X with a kinetic energy of 4.5 J. At point Y, the ball has a speed of 9.5 m s^{-1} . Air resistance is negligible.

- (i) For the movement of the ball from X to Y, draw a solid line on Fig. 2.1 to show:
- the distance moved (label this line D)
 - the displacement (label this line S).
- (ii) By consideration of energy transfer, determine the height h of point Y above the ground.

$$h = \dots\dots\dots \text{ m [3]}$$

- (iii) On Fig. 2.2, sketch the variation of the kinetic energy of the ball with its vertical height above the ground for the movement of the ball from X to Y. Numerical values are not required.

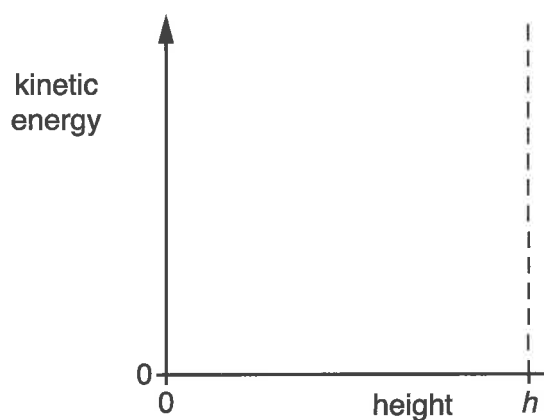


Fig. 2.2

[2]

39. MAR 19/P22/Q2

booklet (scan + OCR)

(a) Define:

(i) *displacement*

..... [1]

(ii) *acceleration*.

..... [1]

- (b) A man wearing a wingsuit glides through the air with a constant velocity of 47 m s^{-1} at an angle of 24° to the horizontal. The path of the man is shown in Fig. 2.1.

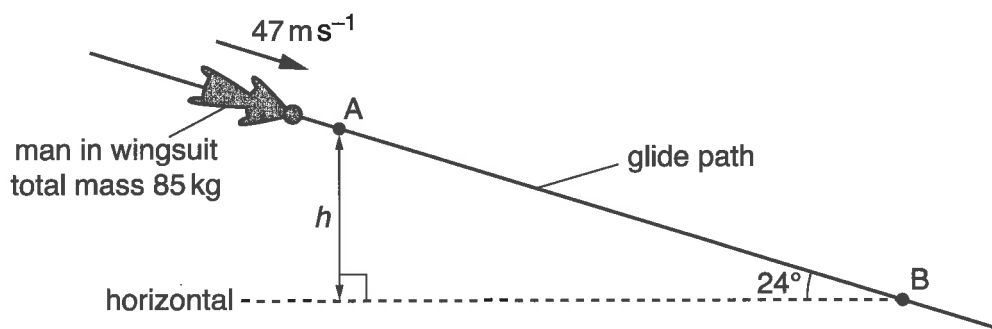


Fig. 2.1 (not to scale)

The total mass of the man and the wingsuit is 85 kg. The man takes a time of 2.8 minutes to glide from point A to point B.

- (i) With reference to the motion of the man, state and explain whether he is in equilibrium.

.....
 [2]

- (ii) Show that the difference in height h between points A and B is 3200 m.

[1]

- (iii) For the movement of the man from A to B, determine:

1. the decrease in gravitational potential energy

decrease in gravitational potential energy = J [2]

2. the magnitude of the force on the man due to air resistance.

force = N [2]

- (iv) The pressure of the still air at A is 63 kPa and at B is 92 kPa. Assume the density of the air is constant between A and B.

Determine the density of the air between A and B.

density = kg m^{-3} [2]

40. O/N 18/P21/Q3

booklet (scan + OCR)

- (a) (i) Define *power*.

..... [1]

- (ii) State what is meant by *gravitational potential energy*.

..... [1]

- (b) An aircraft of mass 1200 kg climbs upwards with a constant velocity of 45 m s^{-1} , as shown in Fig. 3.1.

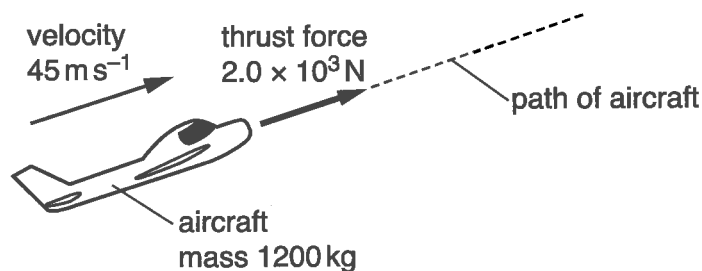


Fig. 3.1 (not to scale)

The aircraft's engine produces a thrust force of $2.0 \times 10^3\text{ N}$ to move the aircraft through the air. The rate of increase in height of the aircraft is 3.3 m s^{-1} .

- (i) Calculate the power produced by the thrust force.

power = W [2]

- (ii) Determine, for a time interval of 3.0 minutes,

1. the work done by the thrust force to move the aircraft,

work done = J [2]

2. the increase in gravitational potential energy of the aircraft,

increase in gravitational potential energy = J [2]

3. the work done against air resistance.

work done = J [1]

- (iii) Use your answer in (b)(ii) part 3 to calculate the force due to air resistance acting on the aircraft.

force = N [1]

- (iv) With reference to the motion of the aircraft, state and explain whether the aircraft is in equilibrium.

.....

..... [2]

41. O/N 18/P23/Q2

booklet (scan + OCR)

(a) State what is meant by *kinetic energy*.

..... [1]

(b) A cannon fires a shell vertically upwards. The shell leaves the cannon with a speed of 80 m s^{-1} and a kinetic energy of 480 J. The shell then rises to a maximum height of 210 m. The effect of air resistance is significant.**(i)** Show that the mass of the shell is 0.15 kg.

[2]

(ii) For the movement of the shell from the cannon to its maximum height, calculate

1. the gain in gravitational potential energy,

gain in gravitational potential energy = J [2]

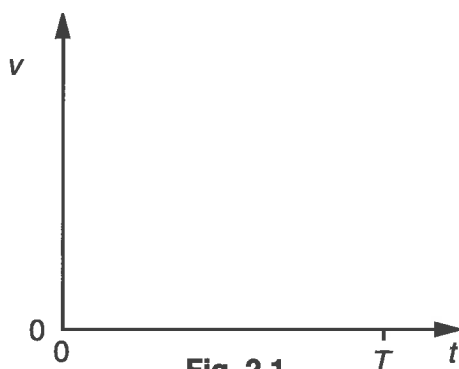
2. the work done against air resistance.

work done = J [1]

(iii) Determine the average force due to the air resistance acting on the shell as it moves from the cannon to its maximum height.

force = N [2]

(iv) The shell leaves the cannon at time $t = 0$ and reaches maximum height at time $t = T$.
On Fig. 2.1, sketch the variation with time t of the velocity v of the shell from time $t = 0$ to time $t = T$. Numerical values of v and t are not required.

**Fig. 2.1****[2]**

(v) The force due to the air resistance is a vector quantity.

Compare the force due to the air resistance acting on the shell as it rises with the force due to the air resistance as it falls.

.....

..... **[2]**

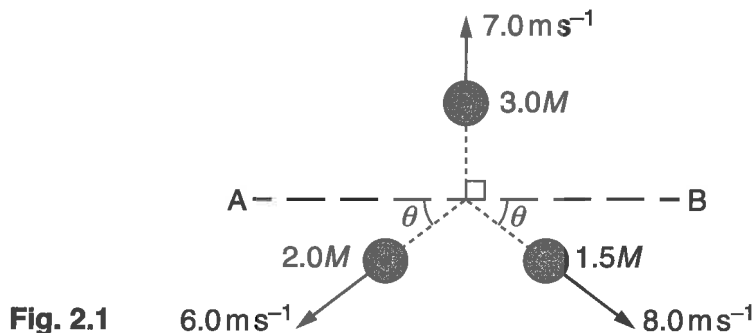
42. M/J 18/P22/Q2

booklet (scan + OCR)

(a) State the principle of conservation of momentum.

.....

..... [2]

(b) A stationary firework explodes into three different fragments that move in a horizontal plane, as illustrated in Fig. 2.1.

The fragment of mass $3.0M$ has a velocity of 7.0 ms^{-1} perpendicular to line AB.
 The fragment of mass $2.0M$ has a velocity of 6.0 ms^{-1} at angle θ to line AB.
 The fragment of mass $1.5M$ has a velocity of 8.0 ms^{-1} at angle θ to line AB.

(i) Use the principle of conservation of momentum to determine θ .
$$\theta = \dots\dots\dots^\circ \text{ [3]}$$
(ii) Calculate the ratio

$$\frac{\text{kinetic energy of fragment of mass } 2.0M}{\text{kinetic energy of fragment of mass } 1.5M}$$

$$\text{ratio} = \dots\dots\dots \text{ [2]}$$
43. M/J 18/P23/Q3

booklet (scan + OCR)

A ball is thrown vertically upwards towards a ceiling and then rebounds, as illustrated in Fig. 3.1.

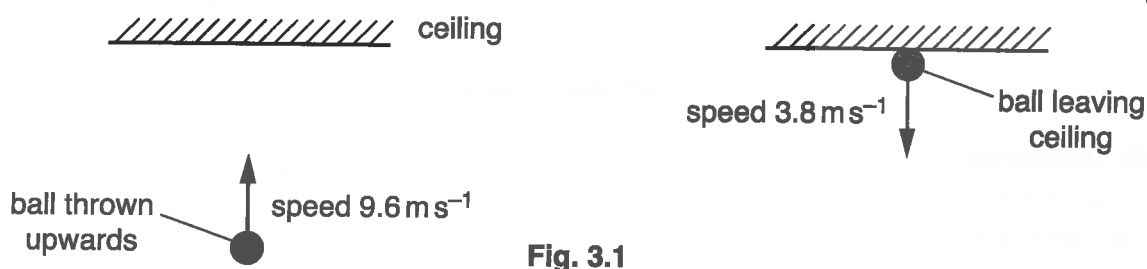


Fig. 3.1

The ball is thrown with speed 9.6 m s^{-1} and takes a time of 0.37 s to reach the ceiling. The ball is then in contact with the ceiling for a further time of 0.085 s until leaving it with a speed of 3.8 m s^{-1} . The mass of the ball is 0.056 kg . Assume that air resistance is negligible.

(a) Show that the ball reaches the ceiling with a speed of 6.0 m s^{-1} .

(b) Calculate the height of the ceiling above the point from which the ball was thrown. [1]

height = m [2]

(c) Calculate

(i) the increase in gravitational potential energy of the ball for its movement from its initial position to the ceiling,

increase in gravitational potential energy = J [2]

(ii) the decrease in kinetic energy of the ball while it is in contact with the ceiling.

decrease in kinetic energy = J [2]

(d) State how Newton's third law applies to the collision between the ball and the ceiling.

.....
 [2]

(e) Calculate the change in momentum of the ball during the collision.

change in momentum = N s [2]

(f) Determine the magnitude of the average force exerted by the ceiling on the ball during the collision.

average force = N [2]

44. MAR 18/P22/Q2

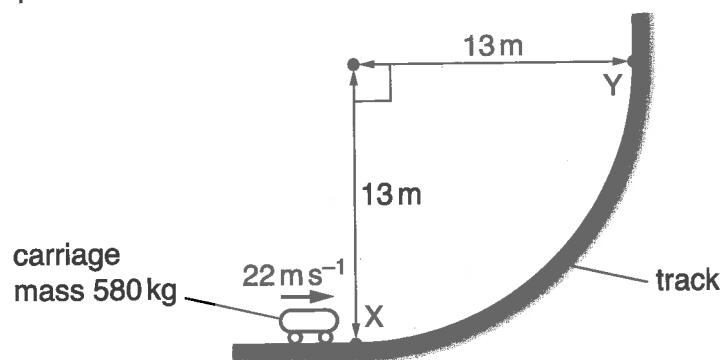
booklet (scan + OCR)

(a) Explain what is meant by**(i)** *work done*,

..... [1]

(ii) *kinetic energy*.

..... [1]

(b) A leisure-park ride consists of a carriage that moves along a railed track. Part of the track lies in a vertical plane and follows an arc XY of a circle of radius 13 m, as shown in Fig. 2.1.**Fig. 2.1**

The mass of the carriage is 580 kg. At point X, the carriage has velocity 22 m s⁻¹ in a horizontal direction. The velocity of the carriage then decreases to 12 m s⁻¹ in a vertical direction at point Y.

(i) For the carriage moving from X to Y**1.** show that the decrease in kinetic energy is $9.9 \times 10^4 \text{ J}$,

[2]

2. calculate the gain in gravitational potential energy.

gain in gravitational potential energy = J [2]

- (ii) Show that the length of the track from X to Y is 20 m.

[1]

- (iii) Use your answers in (b)(i) and (b)(ii) to calculate the average resistive force acting on the carriage as it moves from X to Y.

resistive force = N [2]

- (iv) Describe the change in the direction of the linear momentum of the carriage as it moves from X to Y.

..... [1]

- (v) Determine the magnitude of the change in linear momentum when the carriage moves from X to Y.

change in momentum = N s [3]

45. M/J 16/P23/Q3

booklet (scan + OCR)

- (a) Explain what is meant by *gravitational potential energy* and by *kinetic energy*.

gravitational potential energy:

kinetic energy: [2]

- (b) A motion sensor is used to measure the velocity of a ball falling vertically towards the ground, as illustrated in Fig. 3.1.

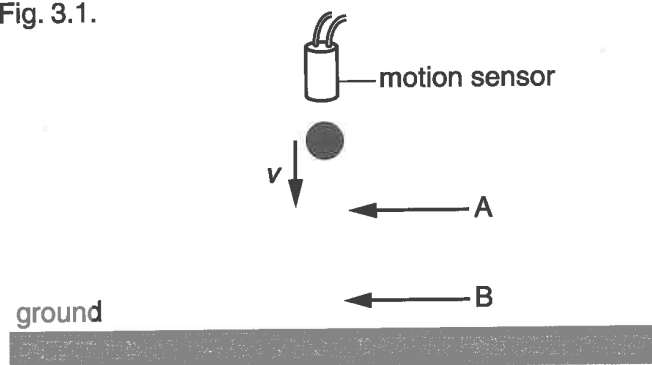


Fig. 3.1

The ball passes through points A and B as it falls. The ball has a mass of 1.5 kg. The variation with time t of the velocity v of the ball as it falls from A to B is shown in Fig. 3.2.

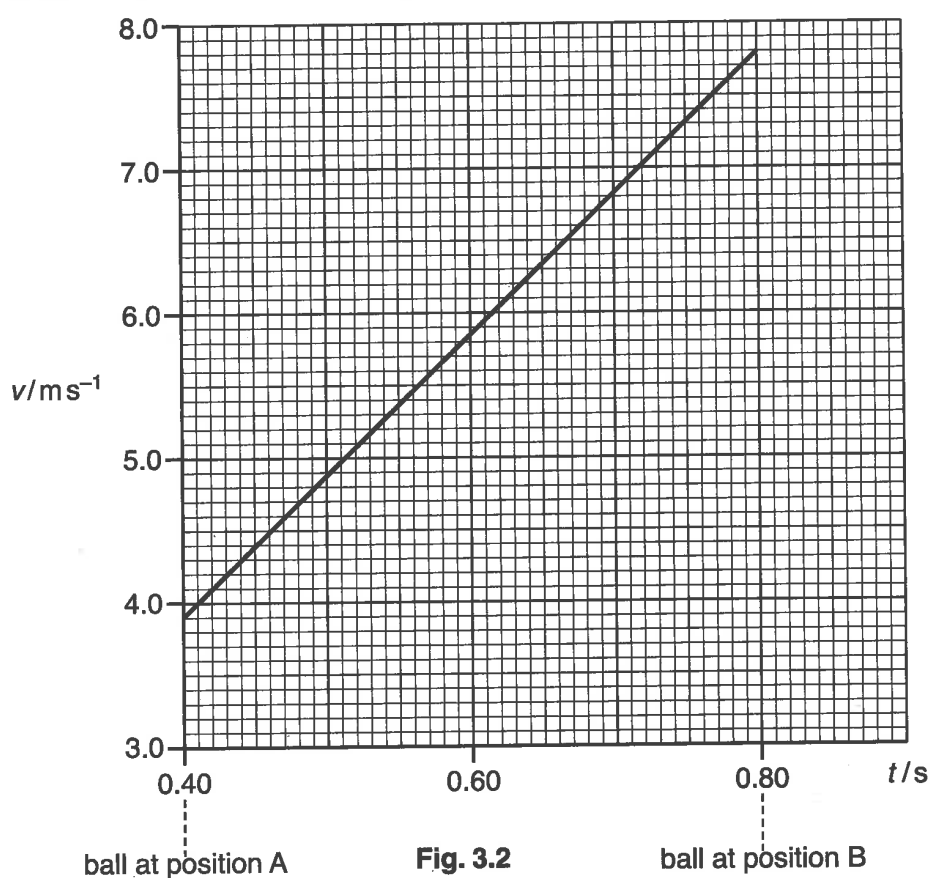


Fig. 3.2

Use Fig. 3.2 to calculate, for the ball falling from A to B,

- (i) the displacement,

displacement = m [3]

(ii) the acceleration,

acceleration = ms^{-2} [2]

(iii) the change in kinetic energy.

change in kinetic energy = J [3]

- (c) Show that the work done by the gravitational field on the ball in (b) as it moves from A to B is equal to the change in kinetic energy.

[2]

46. O/N 15/P23/Q8

booklet (scan + OCR)

- (a) State the quantities, other than momentum, that are conserved in a nuclear reaction.

.....
 [2]

- (b) A stationary nucleus of uranium-238 decays to a nucleus of thorium-234 by emitting an α -particle. The kinetic energy of the α -particle is $6.69 \times 10^{-13} \text{ J}$.

- (i) Show that the kinetic energy E_k of a mass m is related to its momentum p by the equation

$$E_k = \frac{p^2}{2m}.$$

[1]

- (ii) Use the conservation of momentum to determine the kinetic energy, in keV, of the thorium nucleus.

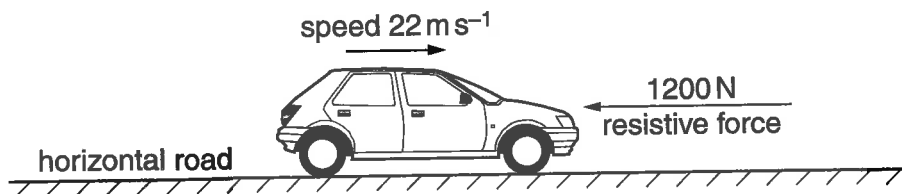
kinetic energy = keV [3]

47. M/J 15/P23/Q3

booklet (scan + OCR)

(a) Define *power*.

..... [1]

(b) Fig. 3.1 shows a car travelling at a speed of 22 m s^{-1} on a horizontal road.**Fig. 3.1**The car has a mass of 1500 kg . A resistive force of 1200 N acts on the car.

Calculate

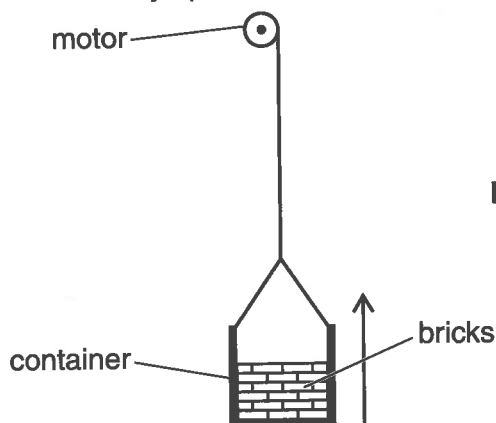
(i) the force F required from the car to produce an acceleration of 0.82 ms^{-2} , $F = \dots\dots\dots \text{ N}$ [3]**(ii)** the power required to produce this acceleration.power = $\dots\dots\dots \text{ W}$ [2]**(c)** The resistive force on the car is proportional to v^2 , where v is the speed of the car.
Suggest why the car has a maximum speed.

..... [1]

48. O/N 14/P21/Q5

booklet (scan + OCR)

A motor is used to move bricks vertically upwards, as shown in Fig. 5.1.

**Fig. 5.1**

The bricks start from rest and accelerate for 2.0s. The bricks then travel at a constant speed of 0.64 m s^{-1} for 25s. Finally the bricks are brought to rest in a further 3.0s.

The total mass of the bricks is 25 kg.

(a) Determine the change in kinetic energy of the bricks

(i) in the first 2.0s,

change in kinetic energy = J [2]

(ii) in the next 25s,

change in kinetic energy = J [1]

(iii) in the final 3.0s.

change in kinetic energy = J [1]

(b) The bricks are in a container. The weight of the container and bricks is 350N.

Calculate, for the lifting of the bricks and container when travelling at constant speed,

(i) the gain in potential energy,

energy gain = J [3]

(ii) the power required.

power = W [2]

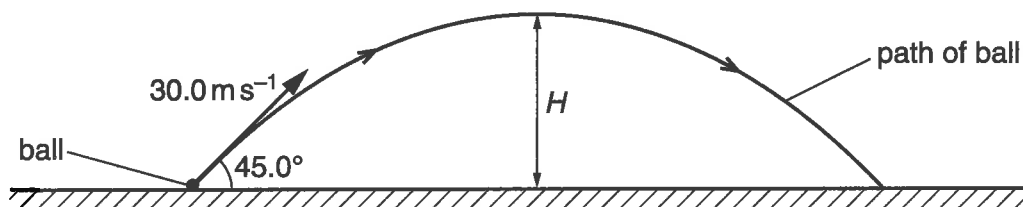
49. M/J 14/P22/Q4

booklet (scan + OCR)

(a) Explain what is meant by *gravitational potential energy* and *kinetic energy*.

gravitational potential energy:

kinetic energy: [2]

(b) A ball of mass 400 g is thrown with an initial velocity of 30.0 ms^{-1} at an angle of 45.0° to the horizontal, as shown in Fig. 4.1.**Fig. 4.1**Air resistance is negligible. The ball reaches a maximum height H after a time of 2.16 s.**(i)** Calculate

1. the initial kinetic energy of the ball,

kinetic energy = J [3]

2. the maximum height
- H
- of the ball,

 $H = \dots\dots\dots \text{m}$ [2]

3. the gravitational potential energy of the ball at height
- H
- .

potential energy = J [2]

(ii) 1. Determine the kinetic energy of the ball at its maximum height.

kinetic energy = J [1]

2. Explain why the kinetic energy of the ball at maximum height is not zero.

..... [1]

50. M/J 14/P21/Q2

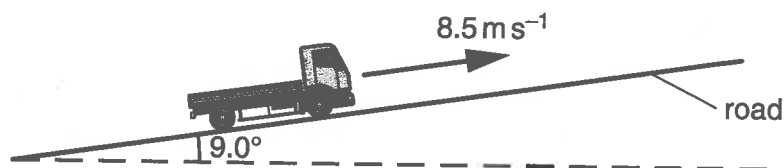
booklet (scan + OCR)

(a) (i) Define *power*.

..... [1]

(ii) Use your definition in **(i)** to show that power may also be expressed as the product of force and velocity.

[2]

(b) A lorry moves up a road that is inclined at 9.0° to the horizontal, as shown in Fig. 2.1.**Fig. 2.1**

The lorry has mass 2500 kg and is travelling at a constant speed of 8.5 m s^{-1} . The force due to air resistance is negligible.

(i) Calculate the useful power from the engine to move the lorry up the road.

power = kW [3]

(ii) State two reasons why the rate of change of potential energy of the lorry is equal to the power calculated in **(i)**.

1.

2. [2]

51. M/J 14/P23/Q3

booklet (scan + OCR)

(a) Explain what is meant by *work done*.

..... [1]

(b) A boy on a board B slides down a slope, as shown in Fig. 3.1.

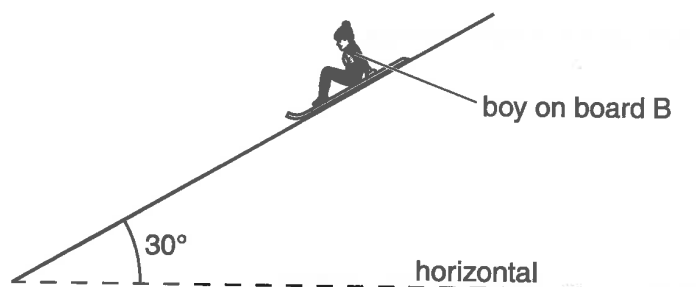


Fig. 3.1

The angle of the slope to the horizontal is 30° . The total resistive force F acting on B is constant.

(i) State a word equation that links the work done by the force F on B to the changes in potential and kinetic energy.

..... [1]

- (ii) The boy on the board B moves with velocity v down the slope. The variation with time t of v is shown in Fig. 3.2.

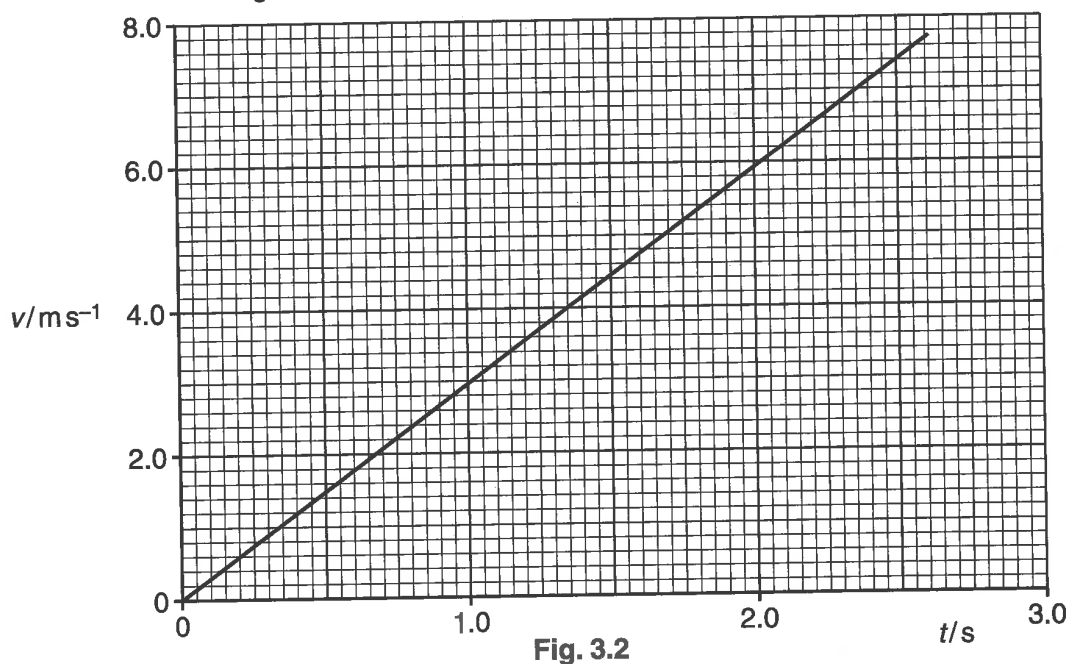


Fig. 3.2

The total mass of B is 75 kg.

For B, from $t = 0$ to $t = 2.5$ s,

- show that the distance moved down the slope is 9.3 m,

[2]

- calculate the gain in kinetic energy,

gain in kinetic energy = J [3]

- calculate the loss in potential energy,

loss in potential energy = J [3]

- calculate the resistive force F .

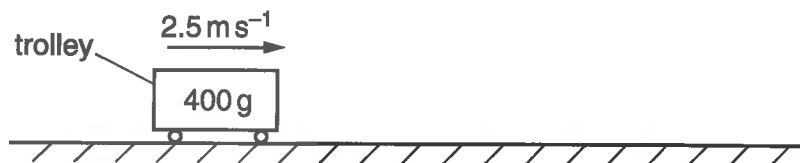
$F =$ N [3]

52. O/N 13/P21/Q3

booklet (scan + OCR)

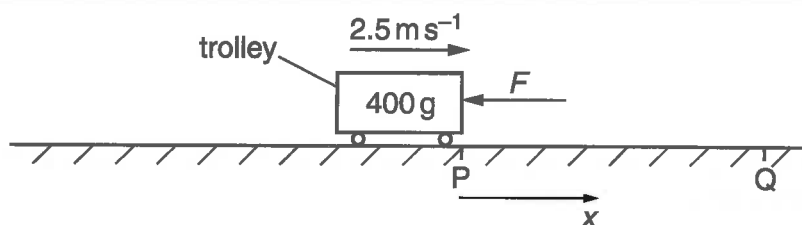
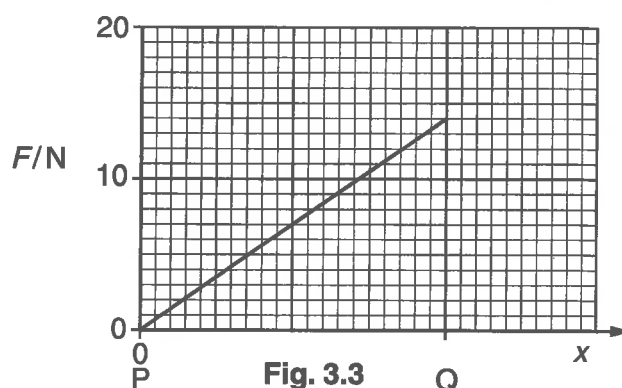
(a) State what is meant by *work done*.

[1]

(b) A trolley of mass 400 g is moving at a constant velocity of 2.5 m s^{-1} to the right as shown in Fig. 3.1.**Fig. 3.1**

Show that the kinetic energy of the trolley is 1.3 J.

[2]

(c) The trolley in **(b)** moves to point P as shown in Fig. 3.2.**Fig. 3.2**At point P the speed of the trolley is 2.5 m s^{-1} .A variable force F acts to the left on the trolley as it moves between points P and Q.The variation of F with displacement x from P is shown in Fig. 3.3.**Fig. 3.3**

The trolley comes to rest at point Q.

(i) Calculate the distance PQ.

distance PQ = m [3]

(ii) On Fig. 3.4, sketch the variation with x of velocity v for the trolley moving between P and Q.

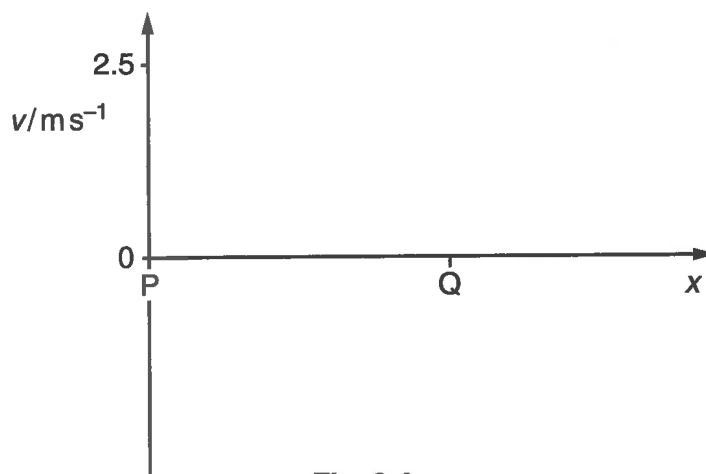


Fig. 3.4

[2]

53. O/N 13/P23/Q4

booklet (scan + OCR)

(a) Distinguish between gravitational potential energy and elastic potential energy.

.....

..... [2]

(b) A ball of mass 65 g is thrown vertically upwards from ground level with a speed of 16 ms^{-1} . Air resistance is negligible.

- (i)** Calculate, for the ball,
1. the initial kinetic energy,

kinetic energy = J [2]

2. the maximum height reached.

maximum height = m [2]

(ii) The ball takes time t to reach maximum height. For time $\frac{t}{2}$ after the ball has been thrown, calculate the ratio

$$\frac{\text{potential energy of ball}}{\text{kinetic energy of ball}}$$

ratio = [3]

(iii) State and explain the effect of air resistance on the time taken for the ball to reach maximum height.

..... [1]

54. M/J 13/P21/Q3

booklet (scan + OCR)

- (a)** An object falls vertically from rest through air. State and explain the energy conversions that occur as the object falls.

.....

.....

..... [3]

- (b)** A ball of mass 150g is thrown vertically upwards with an initial speed of 25 m s^{-1} .

- (i)** Calculate the initial kinetic energy of the ball.

kinetic energy = J [3]

- (ii)** The ball reaches a height of 21 m above the point of release.

For the ball rising to this height, calculate

- 1.** the loss of energy of the ball to air resistance,

energy loss = J [3]

- 2.** the average force due to the air resistance.

force = N [2]

Answers Section

1. M/J 26/P22/Q3/c(i)

3(c)(i)	$(\Delta)E = mg(\Delta)h$	C1	
	$= 0.44 \times 9.81 \times (0.15 \times 2)$	C1	Allow $g = 10$ as leads to same answer to 2SF Allow $0.44 \times 9.81 \times (-0.15 \times 2)$ $0.44 \times 9.81 \times 0.15$ implies 1st C1 mark but not 2nd C1 mark
	$= 1.3 \text{ J}$	A1	Allow -1.3 J

2. M/J 26/P22/Q4/a,b

4(a)	$E_{(K)} = \frac{1}{2}mv^2$	C1	Allow any value of energy equated to $\frac{1}{2}mv^2$
	$v^2 = (2 \times 1.3 \times 10^5 / 1.5 \times 10^4)$	C1	When candidates initially state the correct symbol equation, but then at the substitution stage write v instead of v^2 , treat this as TE in the 2nd mark and allow ECF to the 3rd mark (max 2/3 marks). When candidates do not initially state a symbol equation, and then at the substitution stage write v instead of v^2 , this implies an incorrect symbol equation and so treat as XP (no credit 0/3).
	$v = 4.2 \text{ m s}^{-1}$	A1	
4(b)(i)	work (done) / time (taken)	B1	Allow 'work (done) per (unit) time' Allow 'ratio of work and time' Allow 'rate of (doing) work' Allow 'work over time' (BOD) Not 'work in unit time' Not 'rate of (doing) work per unit time' Not 'change of work (done) per unit time' Ignore 'work (done) by time' Ignore 'work (done) into time' Allow 'energy transferred / time' or 'energy converted / time' Allow 'change in energy / time' (BOD) Allow 'rate of energy transferred' Ignore just 'energy / time' Ignore symbols unless symbols are explained Ignore any units e.g. ignore 'work done per second' or 'Joules per time'
4(b)(ii)	$W = Fs$ or $W = Fvt$ or $W = Pt$ useful work done = change in $E_p + Fs$	C1	Allow $(\Delta)E$ for W and allow 'd' or 'x' for 's' Allow 'WD' for work done (BOD) Allow any value of energy equated to F_s for 1st C1 mark
	$W = (1.1 \times 10^5 \times 52) - (13.0 \times 10^6 - 7.4 \times 10^6)$ $= 1.2 \times 10^5$	C1	Allow F_s or Fvt or Pt for W Treat not converting or incorrect conversion of both 13.0 MJ and 7.4 MJ to J as a single POT error ECF speed from 4(a)
	$F = 1.2 \times 10^5 / (4.2 \times 52)$ $= 550 \text{ N}$	A1	Allow 555 N (correct to 3SF because use of $v=4.16$ gives 554.7 N) but do not allow any other value that rounds to 560
	OR $P = Fv$ useful output power = power to raise weight + Fv	(C1)	Allow any value of power equated to Fv for this mark
	$P = 1.1 \times 10^5 - [(13.0 \times 10^6 - 7.4 \times 10^6) / 52]$ $= 2.3 \times 10^3$	(C1)	Allow Fv for P Treat not converting or incorrect conversion of both 13.0 MJ and 7.4 MJ to J as a single POT error ECF speed from 4(a)
	$F = (2.3 \times 10^3) / 4.2$ $= 550 \text{ N}$	(A1)	Allow 555 N (3SF) but Not 560 N
	OR $P = Fv$ driving force = component of weight parallel to slope + F	(C1)	

4(b)(ii)	$1.1 \times 10^5 / 4.2 = F + 1.5 \times 10^4 \times 9.81 \sin \theta$ <u>and</u> $\sin \theta = (13.0 \times 10^6 - 7.4 \times 10^6) / (1.5 \times 10^4 \times 9.81 \times 4.2 \times 52)$ $1.1 \times 10^5 / 4.2 = F + (13.0 \times 10^6 - 7.4 \times 10^6) / (4.2 \times 52)$	(C1)	ECF speed from 4(a) Treat not converting or incorrect conversion of both 13.0 MJ and 7.4 MJ to J as a single POT error
	$F = 2.619 \times 10^4 - 2.564 \times 10^4$ $= 550 \text{ N}$	(A1)	Allow 555 N (3SF) but not 560 N
4(b)(iii)	$P = IV$ $= 95 \times 1400$ $= 1.33 \times 10^5$	C1	For both methods C1 marks may be credited independently Allow $V = IR$ <u>and</u> either $P = V^2 / R$ or $P = I^2 R$
	efficiency = useful output power / (total) input power ($\times 100$) $= [1.1 \times 10^5 / (95 \times 1400)] (\times 100)$ $= 83\%$	C1	Ignore 'useful output/total input' but allow 'power' to be implied by substituted values of power
	OR	A1	Candidates may use useful output power = $(13.0 \times 10^6 - 7.4 \times 10^6) / 52 = 1.07692 \times 10^5$ so Allow $(1.08 \times 10^5 / (95 \times 1400)) (\times 100)$ to imply both C1 marks (but will lead to incorrect final answer 81%)
	$E = VI t$ $= 95 \times 1400 \times 52$ $= 6.92 \times 10^6$	(C1)	Ignore 'useful output/total input' but allow 'energy' to be implied by substituted values of energy
	efficiency = useful output energy / (total) input energy ($\times 100$) $= [(1.1 \times 10^5 \times 52) / (1400 \times 95 \times 52)] (\times 100)$ $= 83\%$	(A1)	Candidates may use useful output energy = $(13.0 \times 10^6 - 7.4 \times 10^6) = 5.6 \times 10^6$ so Allow $(5.6 \times 10^6 / (95 \times 1400 \times 52)) (\times 100)$ to imply both C1 marks (but will lead to incorrect final answer 81%)

3. M/J 26/P24/Q1/b(i)

1(b)(i)	$(E_{(K)}) = \frac{1}{2}mv^2$ $= \frac{1}{2} \times 1400 \times 15^2$	C1	
	$= 1.6 \times 10^5 \text{ J}$	A1	

4. MAR 26/P22/Q3/a,b

3(a)(i)	$F = W / d$ $= 5.1 \times 10^3 / 23$	C1	
	$= 220 \text{ N}$	A1	
3(a)(ii)	$(\Delta)E_{(P)} = (16.1 \times 10^3 - 1.3 \times 10^3 - 5.1 \times 10^3)$ $= 9.7 \times 10^3$	C1	
	$(\Delta)E_{(P)} = mg(\Delta)h$	C1	
	$\Delta h = (9.7 \times 10^3) / (280 \times 9.81)$ $= 3.5 \text{ m}$	A1	
3(b)	kinetic energy (at Z) is less (than at X) (due to work done by carriage against resistive forces), so the speed is less (at Z than at X)	B1	

5. O/N 25/P21/Q1/c

1(c)(i)	$(\Delta E_P) = mg\Delta h$	C1	
	$= 2.9 \times 10^6 \times 9.81 \times 3200$ $= 9.1 \times 10^{10} \text{ J}$	A1	

1(c)(ii)	$(\Delta E_K) = \frac{1}{2} m \Delta(v^2)$	C1
	$= \frac{1}{2} \times 2.9 \times 10^6 \times 320^2$	A1
	$= 1.5 \times 10^{11} \text{ J}$	
1(c)(iii)	(power =) work done / time	C1
	power = $((1.5 + 0.91) \times 10^{11}) / 20$	A1
	$= 1.2 \times 10^{10} \text{ W}$	
1(b)(i)	acceleration = $320 / 20$ $= 16 \text{ m s}^{-2}$	A1
1(b)(ii)	$s = \frac{1}{2} ((u) + v)t$ or distance = area under graph	C1
	(height) = $\frac{1}{2} \times 20 \times 320 = 3200 \text{ m}$ or 3.2 km	A1

6. O/N 25/P24/Q3/c(i),d

3(c)(i)	$((\Delta)E) = mg(\Delta)h$	C1
	$= 64 \times 9.81 \times 120 = 75\,000 \text{ J}$ or 75 kJ	A1
3(d)	(all) gravitational potential energy has been converted / equal to elastic potential energy (so no kinetic energy)	B1
	kinetic energy is zero so speed is zero	B1

7. M/J 25/P21/Q1/c(ii)

1(c)(ii)	The (total) initial energy is the same (as in part (b))	B1
	change in gravitational potential energy is same, so speed is the same	B1
1(b)(i)	$s = ut + \frac{1}{2}at^2$ and $u = 0$ or $s = \frac{1}{2}at^2$	C1
	$t = \sqrt{(2 \times 14 / 9.81)}$	
	$t = 1.7 \text{ s}$	A1
	OR $v = \sqrt{(0^2 + 2 \times 9.81 \times 14)}$ $v = 17 \text{ (m s}^{-1}\text{)}$ $17 = (0 + 9.81) \times t$ or $17 = 9.81 \times t$ or $14 = \frac{1}{2} \times (0 + 17) \times t$ or $14 = \frac{1}{2} \times 17 \times t$	(C1)
	$t = 1.7 \text{ s}$	(A1)
1(b)(ii)	$s = ut + \frac{1}{2}at^2$	C1
	$u = (3.6 - \frac{1}{2} \times 9.81 \times 1.7^2) / 1.7$	
	$u = (-) 6.2 \text{ m s}^{-1}$	A1
	OR $v = (3.6 + \frac{1}{2} \times 9.81 \times 1.7^2) / 1.7$ $v = 10 \text{ (m s}^{-1}\text{)}$ $10 = u + 9.81 \times 1.7$ or $10^2 = u^2 + (2 \times 9.81 \times 3.6)$ or $3.6 = \frac{1}{2} \times (u + 10) \times 1.7$	(C1)
	$u = (-) 6.2 \text{ m s}^{-1}$	(A1)

8. M/J 25/P21/Q3/a,b,c(ii)

3(a)	work done per unit time	B1
3(b)(i)	$P = F \times v$ $= 1750 \times 35$	C1
	$= 6.1 \times 10^4 \text{ W}$	A1
3(b)(ii)	$W = Fs$ $= 1750 \times 17\,000$	C1
	$= 3.0 \times 10^7 \text{ J}$	A1
	or $W = Pt$ $= 6.1 \times 10^4 \times (17\,000 / 35)$	(C1)
	$= 3.0 \times 10^7 \text{ J}$	(A1)
3(b)(iii)	$P = V \times I$ Power in $= 600 \times I$	C1
	Efficiency $= \frac{\text{useful power output}}{\text{(total) power input}}$	C1
	$0.85 = 6.1 \times 10^4 / (600 \times I)$	A1
	$I = 120 \text{ A}$	
3(c)(ii)	The motor is producing less power (because of gravitational force / conversion of gravitational potential energy to kinetic energy) so the current will be smaller.	B1

9. M/J 25/P22/Q1/b

1(b)(i)	$W = Fs$ or $W = mas$	B1
	$s = v^2 / 2a$ or $a = v^2 / 2s$ or $as = v^2 / 2$	B1
	$W = ma(v^2 / 2a)$ or $W = m(v^2 / 2s)s$ or $W = m(v^2 / 2)$ and (so E_K) $= \frac{1}{2}mv^2$	B1
	OR	(B1)
	$W = Fs$ or $W = mas$	
	$F = mv / t$ and $s = \frac{1}{2}vt$	(B1)
	$W = mv / t \times \frac{1}{2}vt$ and (so E_K) $= \frac{1}{2}mv^2$	(B1)
	OR	(B1)
	$W = Fs$ or $W = mas$	
	$a = v / t$ and $s = \frac{1}{2}vt$	(B1)
	$W = m(v / t)(\frac{1}{2}vt)$ and (so E_K) $= \frac{1}{2}mv^2$	(B1)
	OR	(B1)
	$W = Fs$ or $W = mas$	
	$a = v / t$ and $s = \frac{1}{2}at^2$	(B1)
	$W = m(v / t)(\frac{1}{2} \times (v / t) \times t^2)$ and (so E_K) $= \frac{1}{2}mv^2$	(B1)
1(b)(ii)	kinetic energy $= \frac{1}{2}mv^2$ $= \frac{1}{2} \times 920 \times 17^2$ $= 1.3 \times 10^5 \text{ J}$	A1

1(b)(iii)	$P = W/t$	C1
	$= (4.7 \times 10^4 + 1.3 \times 10^5) / 5.8$	C1
	$= 3.1 \times 10^4 \text{ W}$	A1
1(b)(iv)	(at/after $t = 5.8 \text{ s}$) the kinetic energy (of the car) does not change / work is done only against resistive forces / no work is done to accelerate (the car) so (power output is) less	B1

10. M/J 25/P22/Q4/c

4(c)	$\frac{1}{2} mv^2 = mg(\Delta)h$ $\frac{1}{2} \times 0.25 \times 5.2^2 = 0.25 \times g \times h_1$ $h_1 = 5.2^2 / 2g$ $h_1 = 1.38$	C1
	$\frac{1}{2} \times 0.25 \times 3.6^2 = 0.25 \times g \times h_2$ $h_2 = 3.6^2 / 2g$ $h_2 = 0.66$ $h_2 / h_1 = 0.66 / 1.38$	C1
	ratio = 0.48	A1

11. M/J 25/P23/Q1/c(ii)

1(c)(ii)	The (total) initial energy is greater (than in part (1b))	B1
	change in gravitational potential energy is same, so speed is greater	B1
1(b)	For object A: $s = ut + \frac{1}{2} at^2$ so $t = \sqrt{\frac{2 \times 10}{9.81}}$ $= 1.4$	C1
	Then for object B: $s = ut + \frac{1}{2} at^2$ $h = -(3 \times 1.4) + (0.5 \times 9.81 \times 1.4^2)$ $= 5.7 \text{ m}$	C1
	OR $h = -(3 \times 1.4) + 10$ $= 5.7 \text{ m}$	A1
		(C1)
		(A1)

12. M/J 25/P23/Q3/a,b,c(ii)

3(a)	$W = Fd$	B1
	$P = Fd / t = Fv$ or $P = Fvt / t = Fv$	B1
3(b)(i)	$(\Delta)E_{(P)} = mg(\Delta)h$	C1
	increase of gravitational potential energy of car in 1.0 s $= 1500 \times 9.81 \times 30 \times \sin 6.0 = 46\,000 \text{ J}$	A1

3(b)(ii)	(Power to overcome total resistive forces) = 1600×30 = 48 000 W	C1
	power = 48 000 + 46 000 = 9.4×10^4 W	A1
3(c)(ii)	Mass / weight has increased so (power will) increase	B1

13. M/J 25/P24/Q3/a,b,c

3(a)	$E_{(K)} = \frac{1}{2}mv^2$ $110 = \frac{1}{2} \times 5.5 \times v^2$	C1
	$v = 6.3 \text{ m s}^{-1}$	A1
3(b)	$(\Delta)E_{(P)} / 20 = mg(\Delta)h$ OR $(\Delta)E_{(P)} / 20 = mgx_0$	C1
	$(x_0 =) 20 / 5.5 \times 9.81 = 0.37 \text{ (m)}$ Allow $(\Delta)h$ for x_0	A1
3(c)	$[\max (E_{(P)})] = 110 + 20 = 130 \text{ J}$	A1

14. MAR 25/P22/Q3/a(ii)

3(a)(ii)	$(\Delta)E = \frac{1}{2}m(\Delta)v^2$	C1
	gain in KE = $\frac{1}{2}m(v^2 - u^2)$ = $\frac{1}{2} \times 9400 \times (22^2 - 13^2)$	C1
	= $1.5 \times 10^6 \text{ J}$	A1
	OR $W = Fs$ = $ma \times d$	(C1)
	gain in KE = $9400 \times 0.88 \times 180$	(C1)
	= $1.5 \times 10^6 \text{ J}$	(A1)

15. O/N 24/P21/Q2/b(ii)

2(b)(ii)	kinetic energy = $\frac{1}{2}mv^2$	C1
	= $\frac{1}{2} \times 1800 \times 40^2$	A1
	= $1.4 \times 10^6 \text{ J}$	

16. O/N 24/P21/Q3/a,b,c(i,ii,iii)

3(a)	product of force and displacement in direction of force	B1
3(b)	weight/force = mg , and g identified as acceleration of free fall	B1
	ΔE_P identified as work (done by lifting force/weight), and so = $mg \times \Delta h$	B1
3(c)(i)	$240 \times 150 = 36 \text{ kJ}$	A1
3(c)(ii)	$P = W / t$	C1
	= $36\,000 / 60$	A1
	= 600 W	

3(c)(iii)	efficiency = <u>useful</u> output power / total input power	C1
	= 600 / 900	A1
	= 0.67	

17. O/N 24/P22/Q2/d

2(d)(i)	$F = W / s$	C1
	= 250 / 18	A1
	= 14 N	
2(d)(ii)	any line starting at distance = 0 and a positive non-zero value of kinetic energy	B1
	a straight line from distance = 0 to distance = x with positive gradient	B1
	a straight horizontal line at a non-zero value of kinetic energy starting at distance = x and ending at distance = $x + 18$ m that is continuous with the previous line	B1
2(b)	$F = m\Delta v / \Delta t$	C1
	= $16 \times 0.60 / 1.1$	
	= 8.7 N	A1
2(c)	$x = ut + \frac{1}{2}at^2$	C1
	$x = 0.60 \times 3.7 + \frac{1}{2} \times 0.85 \times 3.7^2$	
	or	
	$v = 0.60 + 0.85 \times 3.7$ (= 3.75 m s ⁻¹)	(C1)
	$x = 3.75 \times 3.7 - 0.5 \times 0.85 \times 3.7^2$	
	or	
	$x = \frac{1}{2} \times (0.60 + 3.75) \times 3.7$	
	or	
	$x = (3.75^2 - 0.60^2) / (2 \times 0.85)$	
	$x = 8.0$ m	A1

18. M/J 24/P22/Q1/b

1(b)(i)	$I = P / A$	C1
	= $750 / (1300 \times 10^{-3})^2$	C1
	= 440 W m ⁻²	A1
1(b)(ii)	percentage uncertainty = $3 + 2 \times (5 / 1300) \times 100$	C1
	= $3 + 2 \times 0.38$	A1
	= ($\pm$) 4%	
1(b)(iii)	efficiency = useful output power / total input power = $(160 / 750) \times 100$ = 21%	A1
1(b)(iv)	area (of the new panel) is less	B1
	input power (of the new panel) is less (than the input power of the original panel) (as intensity is constant)	B1
	(useful power output is unchanged so) efficiency is greater (than the original panel)	B1

19. M/J 24/P22/Q4/b

4(b)(i)	$(\Delta)E_{(P)} = mg(\Delta)h$	C1
	$= 4.5 \times 10^{-2} \times 9.81 \times 8.0 \times 10^{-2} \sin 15^\circ$	C1
	$= 9.1 \times 10^{-3} \text{ J}$	A1
4(b)(ii)	$E_{(K)} = \frac{1}{2}mv^2$	C1
	$(9.3 \times 10^{-2} - 9.1 \times 10^{-3}) = \frac{1}{2} \times 4.5 \times 10^{-2} \times v^2$	C1
	$v = (2 \times 8.4 \times 10^{-2} / 4.5 \times 10^{-2})^{0.5}$ $= 1.9 \text{ m s}^{-1}$	A1

20. M/J 24/P23/Q2/b(v)

2(b)(v)	$KE = \frac{1}{2}mv^2$ or $GPE = mgh$	C1
	initial KE + loss in GPE = final KE	C1
	$(\frac{1}{2} \times m \times 6.0^2) + (m \times 9.81 \times 2.5) = (\frac{1}{2} \times m \times v^2)$	
	$v = 9.2 \text{ m s}^{-1}$	A1

21. MAR 24/P22/Q2/b

2(b)	$\frac{1}{2} m(\Delta)v^2 = mg(\Delta)h$	C1
	$v^2 = 5.9^2 + 2 \times 9.81 \times 7.8$ $v^2 = 188$	C1
	$v = 14 \text{ m s}^{-1}$	A1
	or by resolving components	(C1)
	Vertically: $v^2 = u^2 + 2as$ $v^2 = (5.9 \sin 60)^2 + 2 \times -9.81 \times (1.2 - 9.0)$ $v_v = 13.4$	
	horizontally: $v_h = 5.9 \cos 60$ $v_h = 2.95$	(C1)
	resultant velocity $= \sqrt{(13.4^2 + 2.95^2)}$ $= 14 \text{ m s}^{-1}$	(A1)

22. O/N 23/P22/Q3/b,c

3(b)	$E_{(K)} = \frac{1}{2}mv^2$	C1
	$0.048 = \frac{1}{2} \times 7.5 \times 10^{-3} \times v^2$	A1
	$v = 3.6 \text{ m s}^{-1}$	
3(c)(i)	$(\Delta)E = mg(\Delta)h$	C1
	$0.039 = 7.5 \times 10^{-3} \times 9.81 \times (\Delta)h$	A1
	$\Delta h = 0.53 \text{ m}$	
3(c)(ii)	$F \times 0.53 = 0.048 - 0.039$	C1
	$F = 0.02 \text{ N}$	A1

23. O/N 23/P23/Q3/c(iii)

3(c)(iii)	$E_k = \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times 64 \times 22^2$	A1
	$= 15\,000 \text{ J}$	
	or	
	$E_k = p^2 / 2m$	(C1)
	$= 1400^2 / (2 \times 64)$	(A1)
	$= 15\,000 \text{ J}$	
3(b)(i)	$F = \rho Vg$ $V = 93\,000 / (1.2 \times 9.81)$ $= 7900 \text{ m}^3$	A1
3(b)(ii)	weight $= 93\,000 + 3(.0) \times 10^3$	C1
	$m = (93\,000 + 3.0 \times 10^3) / 9.81$	A1
	$= 9800 \text{ kg}$	

24. M/J 23/P21/Q2/c

(a) (i) $((\Delta)E) = mg(\Delta)h$ **or** $W(\Delta)h$

$$= 0.30 \times 0.090$$

$$= 0.027 \text{ J}$$

2

(ii) $E = \frac{1}{2}mv^2$

$$E = 0.044 - 0.027 (= 0.017)$$

$$v^2 = (2 \times 0.017) / (0.30 / 9.81) \quad (\text{where } v = \text{speed at A})$$

$$v = 1.1 \text{ m s}^{-1}$$

or

$$E = \frac{1}{2}mv^2$$

$$0.044 = \frac{1}{2} \times (0.30 / 9.81) \times v^2 \quad (\text{where } v = \text{speed at B})$$

$$(v^2 = 2.88)$$

$$v^2 = u^2 + 2as \quad (\text{where } u = \text{speed at A})$$

$$2.88 = u^2 + 2 \times 9.81 \times 0.090$$

$$u = 1.1 \text{ m s}^{-1}$$

3

(iii) (gravitational / resultant) force / weight is vertical

1

(iv) straight line with positive gradient starting from non-zero value of v_y at time t_A to a time t_B

1

25. MAR 23/P22/Q3

- (a) resultant force (in any direction) is zero
resultant moment/torque (about any point) is zero 2
- (b) (component =) $17\sin 50^\circ = 13 \text{ (N)}$
or
 $17\cos 40^\circ = 13 \text{ (N)}$ 1
- (c) $(W \times 0.25)$ or (12×0.35) or (13×0.50)
 $(W \times 0.25) + (12 \times 0.35) = (13 \times 0.50)$
 $W = 9.2 \text{ N}$ 2
- (d) $F = 9.2 + 12 - 13$
 $= 8 \text{ N}$ 1
- (e) decrease 1

26. O/N 22/P21/Q3

- (a) (i) work done per unit time 1
(ii) $W = Fs$
 $P = Fs / t$ and (so) $P = Fv$ 2
- (b) $(F =) 130 \times 10^3 / 25 = 5200 \text{ (N)}$ 1
- (c) (i) (component of weight =) $mg \sin \theta$ 1
(ii) F (along slope due to weight) $= 36\,000 \times 9.81 \times \sin 1.4^\circ$
 $(= 8600 \text{ N})$
- (total) $F = 5200 + 36\,000 \times 9.81 \times \sin 1.4^\circ$
 $(= 13\,800 \text{ N})$
- $P = 13\,800 \times 25$
 $= 350 \times 10^3 \text{ (W)}$
 $= 350 \text{ kW}$ 3

27. M/J 22/P22/Q3

- (a) $v^2 = u^2 + 2as$
 $s = 5.6^2 / (2 \times 9.81)$
(max height =) $3.1 + 5.6^2 / (2 \times 9.81) = 4.7 \text{ (m)}$ 2
- (b) $s = ut + \frac{1}{2}at^2$
 $4.7 = \frac{1}{2} \times 9.81 \times t^2$
 $t = 0.98 \text{ s}$ 2
- (c) line drawn from a non-zero speed at $t = 0$ to a greater speed at $t = T$
a single sloping straight line drawn from $t = 0$ to $t = T$
line starts with a positive non-zero value of v and ends with a negative non-zero value of v 3
- (d) acceleration (of the ball) 1
- (e) (i) (magnitudes of accelerations are) equal / same 1
(ii) (speeds are) equal / same 1

28. M/J 22/P21/Q3

- (a) (i) work (done) / time (taken) 1
- (ii) $W = Fx$ or $E = Fx$
- $(P =) Fx / t = Fv$ or $P = Fvt / t = Fv$ 2
- (b) (i) $T = 430 \sin 11^\circ$ or $430 \cos 79^\circ$
- $= 82 \text{ N}$ 2
- (ii) speed $= 56 / 82$
- $= 0.68 \text{ m s}^{-1}$ 1
- (iii) no change in kinetic energy (of block) 1
- (iv) (input power) $= 56 / 0.80$
- $(= 70 \text{ W})$
- time taken $= 1.2 \times 10^3 / 70$
- $= 17 \text{ s}$
- or
- (useful energy) $= 1200 \times 80 / 100$
- $(= 960 \text{ J})$
- time taken $= 960 / 56$
- $= 17 \text{ s}$ 2

29. M/J 22/P23/Q2

- (a) (time =) displacement / velocity
- (time =) $70.0 / 65.0 \cos 4.30^\circ = 1.08 \text{ (s)}$ 2
- (b) $s = ut + \frac{1}{2}at^2$
- $= (65 \times \sin 4.30^\circ \times 1.08) - (0.5 \times 9.81 \times 1.08^2)$
- $s = -0.46 \text{ (m)}$
- height above ground $= 1.66 - 0.46$
- $= 1.2 \text{ m}$ 3
- (c) GPE has decreased
- (total energy conserved so) KE has increased 2

30. M/J 22/P23/Q4

- (a) work (done) = force $\times$ displacement
- (force = mg and distance $= \Delta h$)
- (so) work (done) $= mg\Delta h$ and work $= \Delta E_{(P)}$ (so $\Delta E_{(P)} = mg\Delta h$) 2
- (b) gravitational potential (energy) to heat/thermal (energy) 1
- (c) (i) $P = mg(\Delta)h / (\Delta)t$ or Fv
- $P = (0.60 \times 9.81 \times 1.4) / 4.0$ or $0.60 \times 9.81 \times (1.4 / 4.0)$
- $= 2.1 \text{ W}$ 2
- (ii) $P = I^2R$ or IV or V^2 / R
- $= 0.09^2 \times 47$ or 0.09×4.23 or $4.23^2 / 47$
- $= 0.38 \text{ W}$ 2
- (iii) efficiency $= P_{\text{out}} / P_{\text{in}} (\times 100)$ or $E_{\text{out}} / E_{\text{in}} (\times 100)$
- $= 0.38 / 2.1 (\times 100)$ or $0.38 \times 4.0 / 2.1 \times 4.0 (\times 100)$
- $= 0.18$ or 18% 2

31. MAR 22/P22/Q4

- (a) ratio = $300 / 3200$
= 0.094 1
- (b) $E = \frac{1}{2}mv^2$ or $E \propto v^2$
ratio = $(0.094)^{0.5}$
= 0.31 2
- (c) work (done against frictional force) = $3200 - 300 (=2900)$
length = $2900 / 76$
= 38 m 2
- (d) (i) $E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ and $F = kx$
 $140 = \frac{1}{2} \times 63 \times x^2$ or $140 = \frac{1}{2}Fx$ and $F = 63x$
 $x = 2.1$ m 2
- (ii) percentage efficiency = $(140 / 300) \times 100$
= 47% 1
- (iii) curved line from the origin
gradient of line increases 2

32. O/N 21/P22/Q3

- (a) work (done)/ time (taken) [1]
- (b) (i) zero / 0 J [1]
- (ii) work done = 440×25
= 1.1×10^4 J [1]
- (c) (i) $(\Delta)E_{(P)} = mg(\Delta)h$
 $h = 4.8 \times 10^4 / (1700 \times 9.81)$
= 2.9 m [2]
- (ii) $\theta = \sin^{-1} (2.9 / 25)$
= 6.7° [1]
- (d) work done = $4.8 \times 10^4 + 1.1 \times 10^4 (= 5.9 \times 10^4$ J)
time = $5.9 \times 10^4 / 1.7 \times 10^4$
= 3.5 s [2]

33. M/J 21/P22/Q3

- (a) $(\Delta)E = mg(\Delta)h$ or $W(\Delta)h$
= $330 \times (4.0 - 1.1)$
= 960 J [2]
- (b) (work =) $960 - 540$
(= 420 J)
distance moved = $(960 - 540) / 52$
= 8.1 m [2]
- (c) (i) $E = \frac{1}{2}mv^2$
 $540 = \frac{1}{2} \times (330 / 9.81) \times v^2$
 $v = 5.7$ m s⁻¹ [2]
- (ii) speed = horizontal component of velocity
= $5.7 \times \cos 41^\circ$
= 4.3 m s⁻¹ [2]

34. MAR 21/P22/Q2**(a)** force $\times$ displacement in the direction of the force [1]**(b) (i)** $E = \frac{1}{2}mv^2$

$$(m =) 23 \times 2 / 16^2 = 0.18 \text{ (kg)} \quad [2]$$

(ii) $(\Delta)E = mg(\Delta)h$

$$60 = 0.18 \times 9.81 \times h$$

$$h = 34 \text{ m} \quad [2]$$

(iii) (work done $=$) $60 - 23$
 $= 37 \text{ (J)}$

$$\text{average resistive force} = 37/34$$

$$= 1.1 \text{ N} \quad [2]$$

(c) air resistance (acting on ball) increases
 resultant force (on ball) decreases
 or weight constant and air resistance increases
 acceleration decreases [3]

35. M/J 20/P22/Q3**(a)** (work done $=$) force $\times$ displacement in direction of the force [1]**(b) (i) 1.** $(\Delta)E = mg(\Delta)h$

$$= 0.42 \times 9.81 \times 78$$

$$= 320 \text{ J} \quad [2]$$

2. $E = \frac{1}{2}mv^2$

$$(\Delta)E = \frac{1}{2} \times 0.42 \times 23^2$$

$$= 110 \text{ J} \quad [2]$$

(ii) work done $= 320 - 110 (= 210 \text{ N})$

$$\text{average resistive force} = 210 / 78$$

$$= 2.7 \text{ N} \quad [2]$$

(c) downward sloping line from $(0, g)$ to a non-zero value on the time axisline is curved with a gradient that becomes less negative **and**the line meets t -axis at time $t < T$ [2]**36. MAR 20/P22/Q3**

(a) force $\times$ displacement in the direction of the force [1]

(b) (i) displacement = 4.4×30
 work done = $140 \cos 30^\circ \times 4.4 \times 30$
 $= 1.6 \times 10^4 \text{ J}$ [3]

(ii) $p = F/A$
 $F = 860 - 140 \sin 30^\circ (= 790)$
 $A = 790/2400$
 $= 0.33 \text{ m}^2$ [3]

(iii) $\sigma = F/A$ or $F/\pi r^2$ or $4F/\pi d^2$
 $9.6 \times 10^6 = 4 \times 140/\pi d^2$
 $d = 4.3 \times 10^{-3} \text{ m}$ [2]

(c) $E = \frac{1}{2}Fx$ or $\frac{1}{2}kx^2$ or area under graph
 $(\Delta)E = \frac{1}{2} \times (140 + 210) \times 0.20 \times 10^{-3}$
 or
 $(\Delta)E = (\frac{1}{2} \times 210 \times 0.60 \times 10^{-3}) - (\frac{1}{2} \times 140 \times 0.40 \times 10^{-3})$
 or
 $(\Delta)E = (140 \times 0.20 \times 10^{-3}) + (\frac{1}{2} \times 0.20 \times 10^{-3} \times 70)$
 or
 $(\Delta)E = [\frac{1}{2} \times 3.5 \times 10^5 \times (0.60 \times 10^{-3})^2] - [\frac{1}{2} \times 3.5 \times 10^5 \times (0.40 \times 10^{-3})^2]$
 $\Delta E = 0.035 \text{ J}$ [3]

37. O/N 19/P23/Q3

(a) (i) mass [1]
 (ii) charge [1]

(b) (i) $E = V/d$ or $E = F/q$
 $F = (1.2 \times 10^3 \times 4.2 \times 10^{-9})/3.6 \times 10^{-2}$
 $= 1.4 \times 10^{-4} \text{ N}$ [3]

(ii) $W = mg$
 $= 5.9 \times 10^{-6} \times 9.81$
 resultant force = $1.4 \times 10^{-4} - (5.9 \times 10^{-6} \times 9.81)$
 $a = F/m$
 $a = [1.4 \times 10^{-4} - (5.9 \times 10^{-6} \times 9.81)]/[5.9 \times 10^{-6}] = 14 \text{ ms}^{-2}$ [4]

(iii) 1. $s = ut + \frac{1}{2}at^2$
 $1.8 \times 10^{-2} = \frac{1}{2} \times 14 \times t^2$
 $t = 0.051 \text{ s}$ [2]
 2. $p = 0.75 \times 0.051$
 $= 0.038 \text{ m}$ [1]

38. M/J 19/P23/Q2

- (a) (i) 1. $W = mas$ [1]
 2. $s = (v^2 - u^2) / 2a$ [1]
 (ii) W /work equals energy transferred/gain or change in kinetic energy
 $W (= mas) = ma(v^2 - u^2) / 2a$
 leading to $W = m(v^2 - u^2) / 2$ (so $KE = \frac{1}{2}mv^2$) [2]
- (b) (i) 1. solid curved line drawn from X to Y along path of ball and labelled D
 2. solid straight line drawn from X to Y and labelled S [2]
 (ii) $(\Delta)E = mg(\Delta)h$
 $4.5 = (0.040 \times 9.81 \times h) + (\frac{1}{2} \times 0.040 \times 9.5^2)$
 $h = 6.9 \text{ m}$ [3]
 (iii) line with a negative gradient starting from a non-zero value of kinetic energy when the vertical height is zero straight line ends at a non-zero value of kinetic energy when the vertical height is h [2]

39. MAR 19/P22/Q2

- (a) (i) distance in a specified direction (from a point) [1]
 (ii) change in velocity / time (taken) [1]
 (b) (i) constant velocity so no resultant force
 no resultant force so in equilibrium [2]
 (ii) (difference in height =) $47 \times 2.8 \times 60 \times \sin 24^\circ = 3200 \text{ m}$ [1]
 (iii) 1 $(\Delta)E = mg(\Delta)h$
 $= 85 \times 9.81 \times 3200$
 $= 2.7 \times 10^6 \text{ J}$ [2]
 2 In terms of energy:
 work done $= 2.7 \times 10^6 \text{ J}$
 force $= 2.7 \times 10^6 / (47 \times 2.8 \times 60)$
 $= 340 \text{ N}$
In terms of forces:
 component of weight along path = force due to air resistance
 force $= 85 \times 9.81 \times \sin 24^\circ$
 $= 340 \text{ N}$ [2]
 (iv) $(\Delta)p = \rho g(\Delta)h$
 $(92 - 63) \times 10^3 = \rho \times 9.81 \times 3200$
 $\rho = 0.92 \text{ kg m}^{-3}$ [2]

40. O/N 18/P21/Q3

- (a) (i) work (done)/time (taken) [1]
 (ii) energy of a mass due to its position in a gravitational field [1]
 (b) (i) $P = Fv$
 $= 2.0 \times 10^3 \times 45$
 $= 9.0 \times 10^4 \text{ W}$ [2]
 (ii) 1. $W = (2.0 \times 10^3) \times (45 \times 3.0 \times 60)$ or $W = 9.0 \times 10^4 \times 3.0 \times 60$
 $W = 1.6 \times 10^7 \text{ J}$ [2]
 2. $(\Delta)E_p = mg(\Delta)h$
 $= 1200 \times 9.81 \times 3.3 \times 3.0 \times 60$
 $= 7.0 \times 10^6 \text{ J}$ [2]

3. $W = 1.6 \times 10^7 - 7.0 \times 10^6$
 $= 9.0 \times 10^6 \text{ J}$ [1]
- (iii) force $= (9.0 \times 10^6) / (45 \times 3.0 \times 60)$
 $= 1.1 \times 10^3 \text{ N}$ [1]
- (iv) constant velocity so no resultant force
 no resultant force so in equilibrium [2]

41. O/N 18/P23/Q2

- (a) energy (of a mass/body/object) due to motion/speed/velocity [1]
- (b) (i) $E = \frac{1}{2}mv^2$
 $480 = \frac{1}{2} \times m \times 80^2$ so $m = 0.15 \text{ kg}$ [2]
- (ii) 1. $E = mgh$ or $\Delta E = mg\Delta h$
 $= 0.15 \times 9.81 \times 210$
 $= 310 \text{ J}$ [2]
2. work done $= 480 - 310$
 $= 170 \text{ J}$ [1]
- (iii) work done $= Fs$
 force $= 170/210$
 $= 0.81 \text{ N}$ [2]
- (iv) curved line from positive value on v-axis to (T, 0)
 magnitude of gradient decreases [2]
- (v) as shell rises force decreases and as shell falls force increases
 as shell rises force is downward and as shell falls force is upward
 or
 as shell rises the force decreases and is downward
 as shell falls the force increases and is upward [2]

42. M/J 18/P22/Q2

- (a) sum/total momentum (of a system of bodies) is constant
 or
sum/total momentum before $=$ sum/total momentum after
 for an isolated system or no (resultant) external force [2]
- (b) (i) $(p =) mv$ or $(3.0M \times 7.0)$ or $(2.0M \times 6.0)$ or $(1.5M \times 8.0)$
 $3.0M \times 7.0 = 2.0M \times 6.0 \sin \theta + 1.5M \times 8.0 \sin \theta$
 $\theta = 61^\circ$
 or (vector triangle method)
 momentum vector triangle drawn
 $\theta = 61^\circ$ (2 marks for $\pm 1^\circ$, 1 mark for $\pm 2^\circ$)
 or (use of cosine rule)
 $p = mv$ or $(3.0M \times 7.0)$ or $(2.0M \times 6.0)$ or $(1.5M \times 8.0)$
 $(21M)^2 = (12M)^2 + (12M)^2 - (2 \times 12M \times 12M \times \cos 2\theta)$
 $\theta = 61^\circ$ [3]
- (ii) $(E =) \frac{1}{2}mv^2$
 ratio $= (\frac{1}{2} \times 2.0M \times 6.0^2) / (\frac{1}{2} \times 1.5M \times 8.0^2)$
 $= 0.75$ [2]

43. M/J 18/P23/Q3

(a) $v = u + at$

$$v = 9.6 - (9.81 \times 0.37) = 6.0 \text{ ms}^{-1} \quad [1]$$

(b) $s = \frac{1}{2} \times (9.6 + 6.0) \times 0.37$

or

$$6.0^2 = 9.6^2 - (2 \times 9.81 \times s)$$

or

$$s = (9.6 \times 0.37) - (\frac{1}{2} \times 9.81 \times 0.37^2)$$

or

$$s = (6.0 \times 0.37) + (\frac{1}{2} \times 9.81 \times 0.37^2)$$

$$s = 2.9 \text{ m} \quad [2]$$

(c) (i) $(\Delta)E = mg(\Delta)h$

$$\Delta E = 0.056 \times 9.81 \times 2.9$$

$$= 1.6 \text{ J} \quad [2]$$

(ii) $E = \frac{1}{2}mv^2$

$$\Delta E = \frac{1}{2} \times 0.056 \times (6.0^2 - 3.8^2)$$

$$= 0.60 \text{ J} \quad [2]$$

(d) force on ball (by ceiling) equal to force on ceiling (by ball)

and opposite (in direction) $[2]$

(e) $(p =) mv$ or 0.056×6.0 or 0.056×3.8

change in momentum $= 0.056 \times (6.0 + 3.8)$

$$= 0.55 \text{ N s} \quad [2]$$

(f) resultant force $= 0.55 / 0.085 (= 6.47 \text{ N})$

force by ceiling $= 6.47 - (0.056 \times 9.81)$

$$= 5.9 \text{ N} \quad [2]$$

44. MAR 18/P22/Q2

(a) (i) force $\times$ distance moved in the direction of the force $[1]$

(ii) energy (of a mass/body) due to motion / speed / velocity $[1]$

(b) (i) 1 $E = \frac{1}{2}mv^2$

$$(\Delta)E = \frac{1}{2} \times 580 \times (22^2 - 12^2) = 9.9 \times 10^4 \text{ J} \quad [2]$$

2 $(\Delta)E = mg(\Delta)h$

$$\Delta E = 580 \times 9.81 \times 13$$

$$= 7.4 \times 10^4 \text{ J} \quad [2]$$

(ii) length $= (2\pi \times 13) / 4$ or $(\pi \times 26) / 4$ or $(\pi \times 13) / 2 = 20 \text{ m} \quad [1]$

(iii) work done against resistive force $= 9.9 \times 10^4 - 7.4 \times 10^4$

average resistive force $= (9.9 \times 10^4 - 7.4 \times 10^4) / 20$
$$= 1300 \text{ N} \quad [2]$$

(iv) from horizontal/right to vertical / up or $90^\circ \quad [1]$

(v) $p = mv$ or (580×22) or (580×12)

$$\Delta p = [(580 \times 12)^2 + (580 \times 22)^2]^{0.5}$$

$$= 1.5 \times 10^4 \text{ N s} \quad [3]$$

45. M/J 16/P23/Q3

- (a) (gravitational potential energy is) the energy/ability to do work of a mass that it has or is stored due to its position/height in a gravitational field
 kinetic energy is energy/ability to do work a object/body/mass has due to its speed/velocity/motion/movement [2]

(b) (i) $s = [(u + v)t]/2$ or acceleration = 9.8/9.75 (using gradient)
 $= [(7.8 + 3.9) \times 0.4]/2$ or $s = 3.9 \times 0.4 + \frac{1}{2} \times 9.75 \times (0.4)^2$
 $s = 2.3(4) \text{ m}$ [3]

(ii) $a = (v - u)/t$ or gradient of line
 $= (7.8 - 3.9)/0.4 = 9.8 \text{ (9.75) m s}^{-2}$ (allow $\pm \frac{1}{2}$ small square in readings) [2]

(iii) $\text{KE} = \frac{1}{2}mv^2$
 change in kinetic energy = $\frac{1}{2}mv^2 - \frac{1}{2}mu^2$
 $= \frac{1}{2} \times 1.5 \times (7.8^2 - 3.9^2)$
 $= 34 \text{ (34.22) J}$ [3]

(c) work done = force $\times$ distance (moved) or Fd or Fx or mgh or mgd or mgx
 $= 1.5 \times 9.8 \times 2.3 = 34 \text{ (33.8) J}$ (equals the change in KE) [2]

46. O/N 15/P23/Q8

- (a) mass-energy
 proton number or charge
 nucleon number [2]

(b) (i) $E_k = \frac{1}{2}mv^2$ and $p = mv$ with working leading to
 [via $E_k = \frac{1}{2}m^2v^2/m$ or $\frac{1}{2}m(p/m)^2$]
 to $E_k = \frac{p^2}{2m}$ [1]

(ii) $p = (2E_k m)^{1/2}$ hence $(2[E_k m]_\alpha)^{1/2} = (2[E_k m]_{\text{Th}})^{1/2}$
 $2 \times [E_k]_{\text{Th}} \times 234 = 2 \times 6.69 \times 10^{-13} \times 4$
 $[E_k]_{\text{Th}} = 1.14 \times 10^{-14} \text{ J}$
 $= 71(.5) \text{ keV}$
 or
 calculation of speed of α -particle = $1.42 \times 10^7 \text{ m s}^{-1}$
 calculation of momentum of α -particle/nucleus = $9.43 \times 10^{-20} \text{ N s}$
 $[E_k]_{\text{Th}} = 1.14 \times 10^{-14} \text{ J}$
 $= 71(.5) \text{ keV}$ [3]

47. M/J 15/P23/Q3

- (a) (power =) work done / time (taken) or rate of work done [1]

(b) (i) $F - R = ma$
 $F = 1500 \times 0.82 + 1200$
 $= 2400 \text{ (2430) N}$ [3]

(ii) $P = Fv$
 $= (2430 \times 22) = 53\,000 \text{ (53\,500) W}$ [2]

- (c) (there is maximum power from car and) resistive force = force produced by car hence no acceleration

or

suggestion in terms of power produced by car and power

wasted to overcome resistive force [1]

48. O/N 14/P21/Q5

- (a) (i) change in kinetic energy $= \frac{1}{2}mv^2$
 $= 0.5 \times 25 \times (0.64)^2 = 5.1(2) \text{ J}$ [2]
- (ii) zero [1]
- (iii) $(-)5.1(2) \text{ J}$ [1]
- (b) (i) $PE = mgh$
 $= 350 \times 0.64 \times 25$
 $= 5600 \text{ J}$ [3]
- (If full length used allow 1/3)
- (ii) $P = Fv$ or gain in PE/t , E_p/t or work done $/t$, W/t
 $= 350 \times 0.64$ or $5600 / 25$
 $= 220 (224) \text{ W}$ [2]

49. M/J 14/P22/Q4

- (a) GPE: energy of a mass due to its position in a gravitational field
 KE: energy (a mass has) due to its motion/speed/velocity [2]
- (b) (i) 1. $KE = \frac{1}{2}mv^2$
 $= \frac{1}{2} \times 0.4 \times (30)^2$
 $= 180 \text{ J}$ [3]
2. $s = 0 + \frac{1}{2} \times 9.81 \times (2.16)^2$ or $s = (30 \sin 45^\circ)^2 / (2 \times 9.81)$
 $= 22.88 (22.9) \text{ m}$ $= 22.94 (22.9) \text{ m}$ [2]
3. $GPE = mgh$
 $= 0.4 \times 9.81 \times 22.88 = 89.8 (90) \text{ J}$ [2]
- (ii) 1. $KE = \text{initial KE} - GPE = 180 - 90 = 90 \text{ J}$ [1]
2. (horizontal) velocity is not zero/(object) is still moving/answer explained in terms of conservation of energy [1]

50. M/J 14/P21/Q2

- (a) (i) work (done)/time (taken) [1]
- (ii) work = force $\times$ displacement (in direction of force)
 power = force $\times$ displacement/time (taken) = force $\times$ velocity [2]
- (b) (i) weight = mg
 $P = Fv = 2500 \times 9.81 \times \sin 9^\circ \times 8.5$ (or use $\cos 81^\circ$)
 $= 33 (32.6) \text{ kW}$ [3]
- (ii) no gain or loss of KE
 no work (done) against air resistance [2]

51. M/J 14/P23/Q3

- (a) work done is the product of force and the distance moved in the direction of the force
 or product of force and displacement in the direction of the force [1]
- (b) (i) work done equals the decrease in GPE – gain in KE [1]
- (ii) 1. distance = area under line
 $= (7.4 \times 2.5) / 2 = 9.3 \text{ m} (9.25 \text{ m})$

or

acceleration from graph $a = 7.4/2.5 (= 2.96)$

and equation of motion $(7.4)^2 = 2 \times 2.96 \times s$ gives $s = 9.3$ (9.25) m

[2]

2. kinetic energy $= \frac{1}{2} mv^2$

$$= \frac{1}{2} \times 75 \times (7.4)^2$$

$$= 2100 \text{ J}$$

[3]

3. potential energy $= mgh$

$$h = 9.3 \sin 30^\circ$$

$$\text{PE} = 75 \times 9.81 \times 9.3 \sin 30^\circ = 3400 \text{ J}$$

[3]

4. work done = energy loss

$$R = (3421 - 2054)/9.3$$

$$= 150 \text{ (147) N}$$

[3]

52. O/N 13/P21/Q3

(a) (work =) force $\times$ distance moved / displacement in the direction of the force

OR when a force moves in the direction of the force work is done

[1]

(b) kinetic energy $= \frac{1}{2} mv^2$

$$= \frac{1}{2} \times 0.4 \times (2.5)^2 = 1.25 \text{ J}$$

[2]

(c) (i) area under graph is work done / work done $= \frac{1}{2} Fx$

$$1.25 = (14 \times x) / 2$$

$$x = 0.18 \text{ (0.179) m} \quad [\text{allow } x = 0.19 \text{ m using kinetic energy} = 1.3 \text{ J}]$$

[3]

(ii) smooth curve from $v = 2.5$ at $x = 0$ to $v = 0$ at Q

curve with increasing gradient

[2]

53. O/N 13/P23/Q4

(a) gravitational PE is energy of a mass due to its position in a gravitational field

elastic PE energy stored (in an object) due to (a force) changing its shape / deformation / being compressed / stretched / strained

[2]

(b) (i) 1. kinetic energy $= \frac{1}{2} mv^2$

$$= \frac{1}{2} \times 0.065 \times 16^2 = 8.3(2) \text{ J}$$

[2]

2. $v^2 = 2gh$ OR $\text{PE} = mgh$

$$h = 16^2 / (2 \times 9.81) = 13(05) \text{ m}$$

[2]

(ii) speed at $t = \frac{1}{2}$ total time $= 8 \text{ (ms}^{-1}\text{)}$ or total $t = 1.63$ or $t_{1/2} = 0.815 \text{ s}$

KE is $\frac{1}{4}$ or h at $t_{1/2} = 9.78 \text{ (m)}$

and PE is $\frac{3}{4}$ of max ratio $= 3$ or ratio $= 9.78 / 3.26 = 3$

[3]

(iii) time is less because (average) acceleration is greater OR average force is greater

[1]

54. M/J 13/P21/Q3

(a) loss in potential energy due to decrease in height (as P.E. $= mgh$)

gain in kinetic energy due to increase in speed (as K.E. $= \frac{1}{2} mv^2$)

special case 'as PE decreases KE increases' (1/2)

increase in thermal energy due to work done against air resistance

loss in P.E. equals gain in K.E. and thermal energy

[3]

(b) (i) kinetic energy $= \frac{1}{2} mv^2$

$$= \frac{1}{2} \times 0.150 \times (25)^2$$

$$= 46.875 = 47 \text{ J}$$

[3]

(ii) 1. potential energy ($= mgh$) $= 0.150 \times 9.81 \times 21$
loss = KE $- mgh = 46.875 - (30.9)$
 $= 15.97 = 16 \text{ J}$

[3]

2. work done = 16 J
work done = force $\times$ distance
 $F = 16 / 21 = 0.76 \text{ N}$

[2]

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